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Integrated modeling for the transition pathway of China's power system†

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Accelerating decarbonization of the power system is at the heart of achieving China's carbon neutrality goal and mitigating global climate change. However, deploying multi-terawatts of variable renewable energy (VRE) may result in substantial system volatility. Here, using a temporally and spatially resolved model co-optimizing capacity expansion and system operation throughout the full 8760 hours in a planning year, we show that achieving -550 MtCO_2 per year of negative emissions is feasible for China's power system by 2060 with 6000 GW of VRE, 5800 GW h of energy storage, and 850 MtCO_2 per year of carbon capture and sequestration (CCS), at the marginal carbon abatement cost of 750–1100 yuan per tCO_2 (about 108–157 \$ per tCO_2). Multi-millions of hectares of land areas are necessary to accommodate the TW-scale installation of solar photovoltaic panels, with restricted land policies resulting in a 3.3% increase in electricity costs. System volatility also surges with higher penetration of VRE, represented by increasing variability in hourly marginal demand cost, necessitating firm resources to ensure capacity adequacy. Although these firm sources can earn higher generation revenues in peak hours, capacity compensation amounting to hundreds of yuan per kW (about tens of \$ per kW) per year is still needed. Effective planning and policy formulation are essential to support China's decarbonization effort for its power sector.

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Broader context

Achieving net-zero or even negative emissions for China's power system is essential to its carbon neutrality goal and mitigating global climate change. Power system expansion models are extensively used to analyze pathways for this transition. However, previous studies usually adopt representative hours (tens to hundreds) to simulate power system operation in a year for capacity expansion, which could cause substantial biases, especially when integrating large-scale resource-dependent variable renewable energy (VRE). In this study, we develop a temporally and spatially resolved model co-optimizing capacity expansion and system operation over 8760 hours throughout a full year with a grid-cell level resolution ($25 \text{ km} \times 25 \text{ km}$ at mid-latitudes) for VRE's expansion to fill this modeling gap and explore the transition pathway of China's power system toward 2060. The results show that it is feasible for China's electricity system to achieve negative emissions with large-scale deployment of VRE and negative emissions technologies, while system volatility also increases significantly, necessitating market reforms to ensure capacity adequacy. Using this novel integrated modeling approach that incorporates technology, infrastructure, and policy factors, we demonstrate the challenges and opportunities facing China's energy planners on the road to carbon neutrality. The methods and findings are also of importance for exploring possible strategies to build a sustainable power system and providing policy implications for other major emitting countries pursuing carbon neutrality.

1 Introduction

With China's announced goal of achieving carbon neutrality before 2060 to mitigate global climate change,¹ many studies have proposed plans for profound decarbonization of the

power sector and accelerated electrification of other sectors.^{2–9} In 2023, China's electricity consumption stood at approximately 9.2 petawatt hours (PW h),¹⁰ constituting over 30% of global power demand.¹¹ Despite a significant increase in renewable energy installations surpassing 1500 gigawatts (GW),¹² China's coal-intensive power system emitted around 5.6 gigatons (Gt) of CO_2 in 2023, contributing to more than 40% of the world's electricity-related emissions.¹¹ As the electrification of various sectors advances, establishing net-zero or negative emissions power systems becomes central to China's climate targets, with projections indicating that the country's electricity demand will rise to over 15 PW h per year by

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mid-century.^{3–6,13,14} Although China possesses abundant variable renewable energy (VRE) resources, such as wind and solar,^{15–17} the transition to a 100% VRE system that achieves net-zero or negative emissions is approached with caution.^{18,19} This caution mainly arises from the uneven distribution of VRE resources²⁰ and the mismatch between generation and demand.²¹ Therefore, integrated modeling to understand the carbon abatement pathway and associated strategies is of significant policy relevance, given that the transition necessitates deploying large-scale VRE, energy storage systems, inter-regional transmission infrastructure, and negative emissions technologies (NETs).

Power system expansion models (PSEMs) are extensively used to explore solutions for transitioning energy systems.²² Generally, long-term PSEMs are formulated as linear programming (LP) that integrate a variety of socioeconomic, technical, and meteorological parameters.^{22,23} Modeling zero- or negative-emissions electricity systems, which rely heavily on large-scale deployments of renewable energies, storage, transmission, carbon capture and sequestration (CCS), and/or direct air capture with carbon sequestration (DACCS) technologies, presents significant challenges for large countries like China or the United States. These challenges stem primarily from the complexity of various energy resource limitations and the computational difficulties in solving large-scale LP problems.^{4,23,24} First, continent-scale geographic coverage is necessary to account for the spatiotemporal correlations in weather systems that impact the power supply from wind, solar, and hydropower. Second, full-year temporal coverage (*i.e.*, 8760 hours) and fine temporal resolution (≤ 1 hour) are required to ensure a better presentation of resource adequacy and accurately capture the variability of VRE, storage/transmission operations, thermal unit commitments, and electricity demand.^{4,22,23} Third, resolved capacity expansion resolution (grid cell or dam site level) is critical to expressing the spatial heterogeneity and regional disparities of VRE and hydropower for their integration distance and generation differentiation.^{4,25} Fourth, possible large-scale deployment of NETs also requires an endogenous formulation of carbon sequestration availability and optimal carbon source-sink linkage based on the uneven distribution of carbon sources and sinks.²⁶

Models designed to analyze the massive decarbonization of nationwide power systems usually adopt the two-stage methodology with limited representative hours²⁷ (ranging from tens to hundreds) first to optimize capacity expansion and then the long-chronological hourly operation.^{2,28–30} In the context of power-system expansion, previous research has indicated that inadequate temporal coverage may lead to significant differences in optimal capacity deployment.²² For example, models using the two-stage methodology (or solely capacity expansion with limited temporal coverage) have large variations in predicted VRE installation and carbon abatement cost for China's power system by mid-century, ranging from 500–2300 GW for wind, 250–5000 GW for solar photovoltaic (PV), and about 150–1400 yuan per tCO₂ for carbon abatement cost.^{2,3,31,32} Although carbon capture, transportation, and storage, alongside biomass

energy, are significant for carbon reduction and climate change mitigation,^{6,33} these factors are often simplified without considering resource availability,^{2–4,28,29} or using a soft-link approach between carbon source-sink matching models and PSEM.^{30,34} Therefore, a refined dynamic high spatiotemporal resolution and explicit representation of the carbon source-sink matching model is needed to address several important inter-related questions: Is it feasible for China's power system to achieve negative emissions? If so, what is the pathway to achieve it? And what are the associated costs?

In this study, we develop a temporally and spatially resolved PSEM, the China Integrated Sustainable Power-system Optimization (CISPO) model, with hourly resolution over a full year (8760 hours) to offer a detailed pathway for the transition of China's electricity system with emissions cap constraints adopted from economy-wide models,^{6,35} using 2022 as the base year and optimizing from 2030 to 2060 (10 years as a step). The CISPO model fundamentally improves the modeling approach and energy-related data integration to better understand China's power-system transition pathway compared to a recent precedent, the RESPO model.⁴ First, the CISPO co-optimizes the expansion for all the generation capacity (compared to VRE only in the RESPO) and hourly dispatch operation (compared to a simplified layer model in the RESPO) subject to projected hourly demand balance, generation output constraints, and emissions constraints. Second, hydropower operation is optimized further at the dam site level (compared to the provincial level in the RESPO). Third, the CISPO explicitly includes carbon sources (thermal power with capture, or direct air capture) and sinks (deep saline aquifer sequestration) and endogenously optimizes the match through transportation routes between any provinces (not considered in the RESPO). Combining fine-assessed energy resource potential and the high-resolution optimization model, we reveal many facets of challenges and opportunities facing China's energy planners on the road to carbon neutrality, including siting large-scale wind, solar, and hydropower facilities, deploying technologies to achieve capture and sequestration of hundreds of megatons (Mt) of CO₂, and introducing capacity payments to firm power to regulate the power system.

2 Results

2.1 Power system expansions toward 2060

Fig. 1 shows the optimized power system expansion toward 2060 in the base scenario, including capacity mix, generation mix, system cost of electricity (SCOE, defined as the total annualized capital expenditure, and operational costs of generation, storage, transmission, and other technologies divided by the yearly system-wide demand; distribution and administration costs are not included²³). We find a negative-emissions electricity system for China is feasible and will not increase the SCOE compared to the 2022 level under our cost assumptions; see the ESI,[†] Section S3 for details. Notably, more stringent emissions constraints necessitate larger-scale deployment of

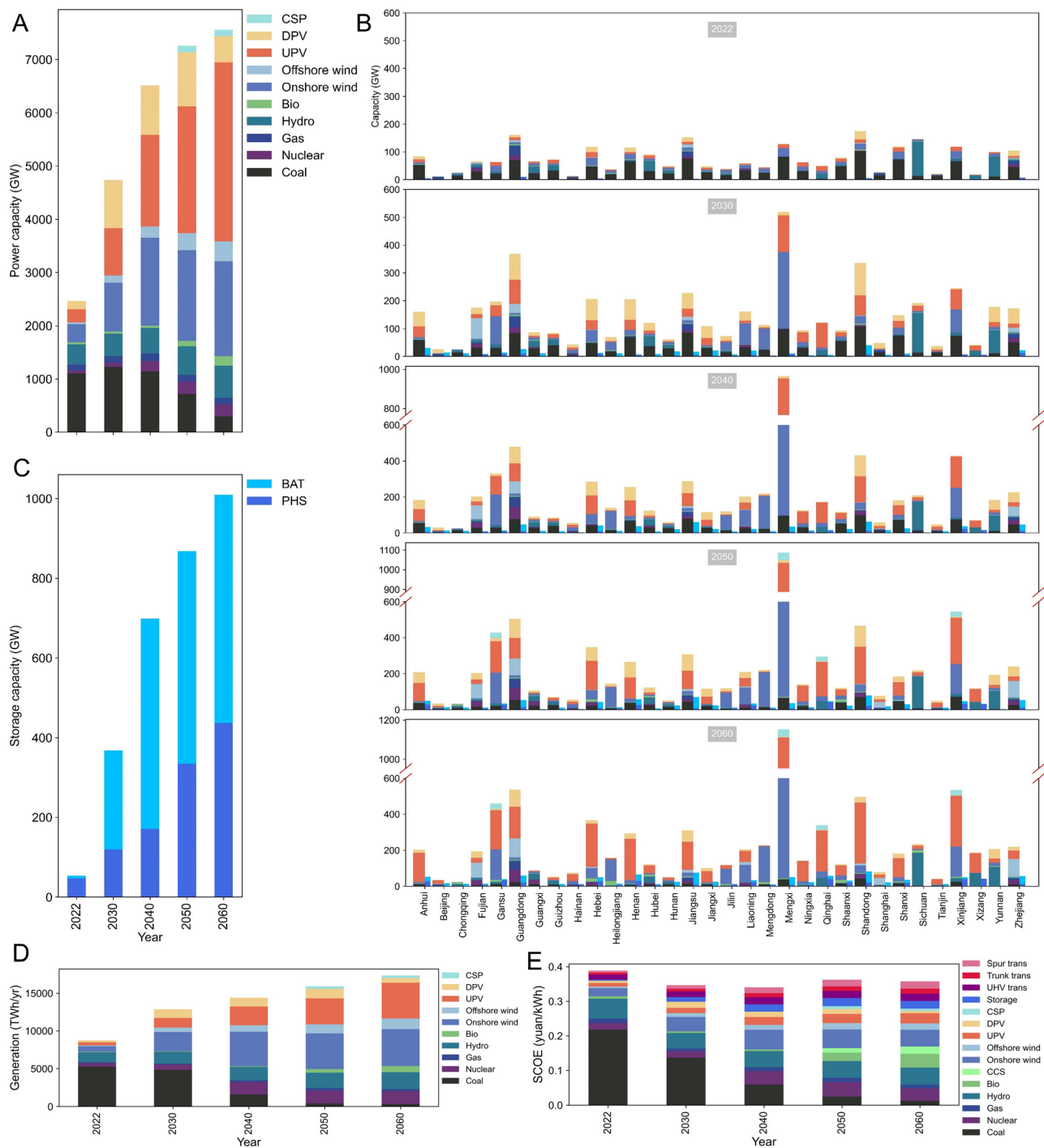


Fig. 1 Optimized results for China's power system toward 2060. (A) Projected capacity (GW) mix of generators. (B) Provincial capacity (GW) breakdown of generator and storage, with Inner Mongolia divided into western (Mengxi) and eastern (Mengdong) regions because they are separate balancing regions. (C) Optimized deployment (GW) of PHS and BAT. (D) Electricity generation mix (TWh per year). (E) System cost of electricity (SCOE, yuan per kWh), excluding distribution and administration costs.

biomass energy with carbon capture and sequestration (BECCS), driving a substantial SCOE increase in later periods.

Accelerated deployment of renewable energy facilitates the decarbonization and transition of China's power system. We find installed capacities of 2.1 terawatts (TW) from wind energy and 3.8 TW from solar photovoltaics (PV) in 2060, increasing

from values of around 1.0 TW and 1.8 TW, respectively, in 2030. These wind and solar PV installations generate, respectively, about 6.3 and 5.3 PW h per year of electricity, contributing to about 72% of total annual generation in 2060. Concentrating solar power (CSP) systems, modeled with an 8-hour energy storage capacity, are projected to reach an installed capacity

of approximately 127 GW and generate 332 TW h annually in 2060. The bulk of these CSP installations (ESI,[†] Fig. S64) are situated in the provinces of Mengxi (40 GW, 100 TW h), Gansu (31 GW, 79 TW h), Xinjiang (30 GW, 83 TW h), and Qinghai (26 GW, 70 TW h), owing to their favorable solar resource endowment and land availability.

Hydropower, a longstanding renewable energy source, has undergone rapid expansion in China over the past few decades, achieving a total installed capacity of 367 GW (excluding pumped storage) by 2022.³⁶ This study optimizes hydropower capacity expansion and reservoir operation at the dam site level, based on a comprehensive assessment of the economic potential of China's hydropower, estimated at approximately 700 GW in total. Our findings suggest that by 2060, the installed capacity of hydropower will reach 600 GW, generating approximately 2.2 PW h per year of electricity. This capacity is predominantly concentrated in China's southwestern provinces, with the top three being Sichuan (182 GW), Yunnan (108 GW), and Xizang (73 GW), accounting for 60% of the installed capacity. See the ESI,[†] Fig. S66, for hydropower's spatial expansion from 2030 to 2060.

Energy storage has proven beneficial in managing the uncertainty and variability in power system arising from the large-scale deployment of VRE sources.^{37,38} In this study, we model two representative storage technologies: pumped hydro storage (PHS) with an 8-hour duration, and lithium-ion batteries (BAT) with a 4-hour duration. By 2060, our model indicates a deployment of 437 GW (3496 GW h) from PHS, and 572 GW (2288 GW h) from BAT (Fig. 1). In this analysis, we use the values reported in the latest PHS resource assessment³⁹ to model PHS's deployment potential, which is constrained by the geographic availability.³⁷ Results show that the leading regions for PHS installation in 2060 are Xizang (42 GW), Qinghai (40 GW), Xinjiang (39 GW), and Gansu (33 GW). In contrast, battery storage sees large-scale capacity installations in Jiangsu (68 GW), Shandong (62 GW), Zhejiang (44 GW), and Guangdong (40 GW), where substantial VRE capacities exist, but lack sufficient PHS potential. In 2030, we observe installations of 120 GW PHS and 248 GW BAT, primarily located in Shandong (8 GW PHS and 33 GW BAT), Anhui (6 GW PHS and 24 GW BAT), and Guangdong (10 GW PHS and 16 GW BAT).

Firm power sources have been shown to play an important role in decarbonized power systems by meeting peak demand, providing inertia and spinning reserve requirements, and enhancing the system's stability to integrate more VRE.^{3,4,28,40} In the CISPO model, operational and under-construction traditional coal and natural gas power plants that existed in the base year will be phased out upon reaching their designed lifetime, or retrofitted with carbon capture equipment to transform into firm low-carbon power sources. Existing biomass, nuclear, and combined heat and power (CHP) units will undergo refurbishment as needed to extend their operational lifetimes rather than retiring. For new builds of thermal power, the model will endogenously optimize the installed capacity along with determining whether CCS implementation is warranted. Simulated results by the model show that there are no new capacities

added for traditional coal and gas power; see the detailed capacity retirement from 2030 to 2060 in the ESI,[†] Section S3.5. In 2060, we find an installed capacity of 181 GW for bioenergy (168 GW with CCS), with 297 GW from coal (including CHP units) and 120 GW from gas (including CHP units), generating about 1.42 PW h per year of electricity.

The most uncertain factor for the long-term nuclear development plan in China is whether nuclear reactors can be deployed inland. We assume newly built nuclear power plants will only be located in the coastal provinces, with the installation capacity potential reported by Xiao and Jiang⁴¹ (about an additional 170 GW) and moderate flexibility (minimum outputs of 85%).^{4,23} These nuclear power plants will be fully installed by 2050, with the nuclear capacity potentials totaling 230 GW when including the capacity from existing and currently under-construction nuclear power plants. This highlights the vital role of nuclear power in achieving a low-carbon electricity system by providing 1.7 PW h of annual generation thereafter. The annual capacity factor of nuclear power peaks in 2040 at 0.96 and declines to 0.85 in 2050 and 2060 with the increased integration of VRE, a higher CCS generation requirement for meeting carbon emissions limits, and the relatively inflexible nature of nuclear plants.

2.2 Emissions and carbon source-sink matching in power

Given the technologies modeled here, a negative emissions power system for China around 2060 will increase system costs by an overall 7–19% (Fig. 2) in scenarios with emissions control compared to the “NoEmisCap” scenario, where carbon emissions constraints are not imposed. We adopt several exogenous emissions pathways from economy-wide studies^{6,35} for the base scenario (−550 MtCO₂ per year), “NegEmis200Mt” (−200 MtCO₂ per year), and “NegEmis700Mt” (−700 MtCO₂ per year) toward 2060 (see Fig. 3). If no emissions constraints are in place, the total emissions from the electricity sector will decrease from approximately 4.5 Gt per year in 2030 to 0.8 Gt

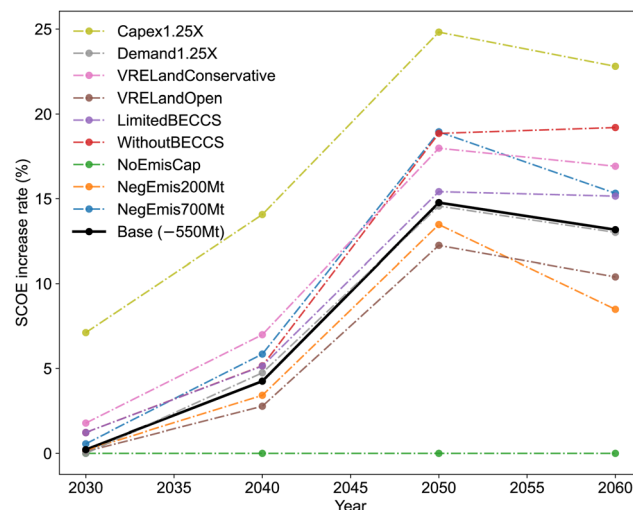


Fig. 2 Increase in system cost of electricity (SCOE, %) compared to the “NoEmisCap” scenario.

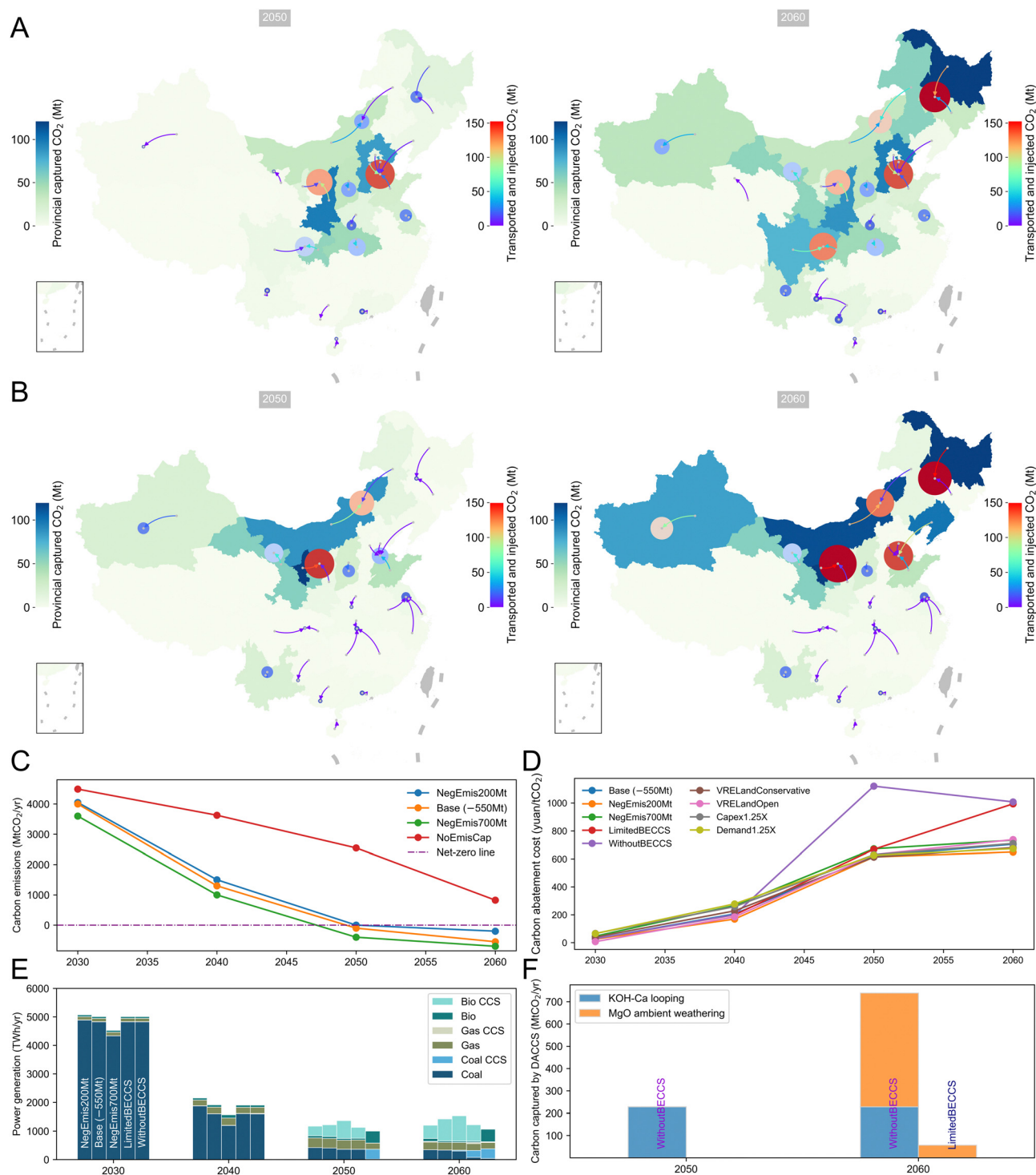


Fig. 3 Results of carbon capture and sequestration deployment, carbon emissions, and abatement costs. (A): Base scenario) and (B): "WithoutBECCS" scenario): Annual carbon (MtCO₂ per year) captured and injected in each province and CO₂ sequestration site, and transportation volume of each capture-sequestration link in 2050 and 2060. Provincial polygons with blue color bar (left) are annual CO₂ captured from thermal plants or DAC. Circles indicate annual CO₂ injected into each sequestration site. Arrows with shades of color represent the carbon source-sink route and CO₂ transported along this route (right rainbow color bar). (C) Annual carbon emissions from China's power sector in emissions-related scenarios. (D) Marginal carbon abatement cost (yuan per tCO₂). (E) Annual generation (TWh per year) from thermal generation. (F) Carbon captured (MtCO₂ per year) by DAC in 2050 and 2060 for the "WithoutBECCS" and "LimitedBECCS" scenarios.

per year in 2060, primarily driven by VRE deployment. Across the emissions control scenarios, the highest system cost increase (+19%) occurs in 2050 under the "NegEmis700Mt"

scenario, equivalent to about 900 billion yuan per year (about 128 billion \$ per year), compared to the "NoEmisCap" scenario. This significant cost increase is due to the deployment of 126

GW BECCS, which is required to meet the -400 MtCO_2 per year emissions target (see Fig. 2 and the ESI,[†] Fig. S48, for SCOE breakdowns). More stringent/lenient emissions pathways (“NegEmis700Mt”/“NegEmis200Mt”) see similar wind ($-1.0\%/+2.8\%$) and solar PV ($-0.7\%/+3.0\%$) installed capacities in 2060, compared to the base scenario. Even in the “NoEmisCap” scenario, 2.1 TW installed capacity from wind energy and 3.6 TW installed capacity from solar PV energy are observed in 2060, showing the robustness of large-scale VRE deployment in the later period of China’s pathway to carbon neutrality. Before 2060, there are notable VRE installed capacity differences among emissions-related scenarios. For example, the installed capacity of wind and solar PV energy is 4692 GW in the “NegEmis700Mt” scenario, while 3300 GW is seen in the “NoEmisCap” scenario in 2040 (see the ESI,[†] Fig. S60).

Negative emissions technologies are expected to have significant carbon dioxide removal potential for limiting global warming.⁴² In the CISPO model, NETs include bioenergy with carbon capture and sequestration (BECCS) and direct air capture with carbon sequestration (DACCS), which are the most prominent NETs.⁴² In this study, we assume the biofuel combusted for BECCS in power systems could only be from residues and energy crops on abandoned land to avoid its possible side-effects,⁴² with the resource potential at the grid-cell level obtained from Wang *et al.*⁴³ For optimization of this model, DACCS is integrated with four technologies, including KOH absorption paired with regeneration *via* calcium looping (KOH-Ca looping), KOH absorption paired with regeneration *via* bipolar membrane electrodialysis (BPMED), solid sorbent direct air capture using temperature vacuum swing adsorption, and MgO ambient weathering with regeneration *via* calcination, with cost projection and key parameters obtained from Young *et al.*⁴⁴ The carbon captured from thermal power plants and the air is endogenously optimized to be transported and sequestered to deep saline aquifer (DSA) carbon sequestration sites, with cost assumptions obtained from Wang *et al.*⁴⁵ (see the ESI,[†] Sections S3.8 and S3.9 for input assumptions, and Section S4.3 for the model formula of NETs in the CISPO model).

If available, BECCS is selected by the CISPO to achieve negative emissions around the mid-century. In the base scenario, we find a large-scale deployment of BECCS from 2050, with a capacity of 81 GW and a generation of 409 TW h per year (Fig. 3E). This BECCS generation brings about negative emissions of 495 Mt annually. By 2060, an additional 87 GW of installed capacity and 283 TW h per year electricity generation for BECCS (totaling 168 GW and 773 TW h per year) is required to offset around -850 MtCO_2 per year of emissions, based on an emissions factor assumption of $-1.1 \text{ kg kW}^{-1} \text{ h}^{-1}$ for BECCS.³⁴ More stringent/lenient emissions constraints result in higher/lower BECCS installed capacity and electricity generation. In the “NegEmis700Mt” scenario, BECCS installed capacity and electricity generation increase to 126 GW (+56%) and 629 TW h per year (+54%), respectively, in 2050, while in the “NegEmis200Mt” scenario, they decrease to 67 GW (−17%) and 338 TW h per year (−17%), respectively. In 2060, these

figures increase to 191 GW (+14%) and 891 TW h per year (+15%), and decrease to 102 GW (−39%) and 462 TW h per year (−40%), respectively. More stringent emissions targets after 2050 (“NegEmis700Mt”) may lead to a slight reduction in renewable energy installed capacity, as more electricity generation from BECCS must be integrated, resulting in decreases of 115 GW installed capacity and 216 TW h per year electricity generation for wind and solar (including CSP) power in 2050, and decreases of 72 GW and 166 TW h per year in 2060 (see the ESI,[†] Fig. S49 and S50).

We also consider a scenario that limits the biomass supply (“LimitedBECCS” with annual biofuel potential only one-fourth of the base scenario) and find that the system cost increases slightly in 2060 (+2%) for the 50 MtCO_2 deployment of DACCS to compensate for the installation gap (−40 GW) of BECCS (see Fig. 3E and F). We further consider an extreme scenario that excludes BECCS (“WithoutBECCS” scenario). Results show that about 228 MtCO_2 per year and 779 MtCO_2 per year of DACCS are necessary in 2050 and 2060, respectively. Based on the cost projections from Young *et al.*,⁴⁴ we find carbon-capture figures of 228 MtCO_2 per year with KOH-Ca looping and 551 MtCO_2 per year with MgO ambient weathering for 2060 (Fig. 3F). These DACCS installations increase the system costs by 3.6% in 2050 and 5.3% in 2060, as shown in Fig. 2.

We employ the marginal abatement cost of CO_2 emissions (MAC, yuan per tCO_2) as the primary metric for assessing the cost of achieving different emissions targets for China’s power system. The MAC, or the shadow price of the annual carbon emissions constraint, represents the additional cost required to abate one extra unit of carbon emissions. In the base scenario, the MAC shows a significant surge between 2040 and 2050, rising by approximately 350–450 yuan per tCO_2 from 206 yuan per tCO_2 in 2040 to 622 yuan per tCO_2 in 2050. In contrast, the increase is 150–200 yuan per tCO_2 between 2030 and 2040, and 30–80 yuan per tCO_2 between 2050 and 2060 (see Fig. 3D). Before 2050, abating carbon emissions is primarily achieved through increasing carbon-free generation to substitute for the corresponding thermal power generation. From 2050 onwards, attaining net-zero or negative emissions necessitates investment in NETs, resulting in a substantial increase in the MAC. In the “WithoutBECCS” scenario, the MAC increases by 80% to 1120 yuan per tCO_2 in 2050 compared to the base scenario, and 43% to 1008 yuan per tCO_2 in 2060 for deploying DACCS as a NET. When BECCS is modeled with a restricted biofuel availability (“LimitedBECCS”), the MACs before 2050 are close to the base scenario. However, this value also increases by 43% due to additional DACCS installations in 2060.

Fig. 3A presents the optimal transportation routes for injecting the captured 497 and 853 MtCO_2 into sequestration sites in 2050 and 2060, respectively, in the base scenario. The captured CO_2 is primarily transported to and stored in the Songliao Basin, Bohai Bay Basin, Jiangnan Basin, and Sichuan Basin due to their abundant surrounding biofuel resources and DSA sequestration potential. These carbon sinks collectively accommodate 45% and 58% of the CO_2 captured in 2050 and 2060. Within the CISPO model, the electricity consumed by direct air

carbon capture for operations (e.g., heating) is assumed to be sourced from the provincial grid. Unlike BECCS, the deployment and operation of DACCS are not constrained by fuel resource availability, but are more cost-effective in regions with lower electricity generation costs. In the absence of BECCS as a NET, our findings indicate that DACCS predominantly occurs in Mengxi, Ningxia, Xinjiang, and Liaoning, with captured CO₂ transported to nearby sequestration sites (Fig. 3B). Results for other scenarios are presented in the ESI,[†] Fig. S52–S59. We find that Heilongjiang (−120 MtCO₂ per year), Hebei (−90 MtCO₂ per year), Shaanxi (−75 MtCO₂ per year), and Sichuan (−71 MtCO₂ per year) contribute the most significant negative carbon emissions from the electricity sector in 2060 (ESI,[†] Fig. S51). While the CISPO model allows for retrofitting conventional thermal power plants with carbon capture equipment, the results demonstrate that the retrofitted capacity is minimal (<1%). This limited retrofitting is primarily attributed to the positive residual emissions associated with this technology (90% capture rate⁴⁵), which result in a lower annual capacity factor and consequently higher abatement costs compared to BECCS.

2.3 Spatial siting and land-use requirement of VRE

Cell-level information of VRE sources incorporated into the CISPO model includes the installation capacity potential and hourly available output over a year, which are related to suitable area and meteorological factors (the year 2019 is used in the base scenario), for the spatial modeling (0.25° × 0.25°) of optimal capacity deployment, electricity supply, and integration with load centers.^{4,23} In this study, we first consider different land-use types, available rooftop areas, and technical limitations to estimate the suitable areas (km²) for VRE deployment^{4,16} and then assess the installation capacity potential using density (MW km^{−2}) for each VRE type. Integration of wind (onshore and offshore) and utility-scale solar PV is modeled by first connecting to a substation and then to a load center based on the minimum connection distance rule following Brown and Botterud,²³ and Zhang *et al.*⁴ See the ESI,[†] Sections S2.1 and S3.3 for details. Within the context of this optimization technique, we observe distinct expansion patterns for wind and solar PV installations from 2030 to 2060, and higher capacity of distributed solar PV than that of utility-scale solar PV in 2030, as shown in Fig. 4A and B. See the ESI,[†] Fig. S62–S64, for VRE's spatial expansion in detail.

Wind power installations are rapidly expanding across regions rich in wind resources, which roughly exhibit two nearly parallel belts. The first belt, located in North, Northeast, and Northwest China, possesses substantial potential for onshore wind, accounting for approximately 75% of the national wind installed capacity and electricity generation (see Fig. 4C and D). The highest wind installation factor, defined as the ratio of planned capacity to the installation capacity potential, is found in Mengdong and Mengxi for 2060, with a value of 0.41 (see Fig. 4E). In other provinces of these regions (North, Northeast, and Northwest China), installation factors range between 0.2 and 0.3, indicating significant

untapped wind potential. The second belt, situated along the eastern and southern coastal areas including Jiangsu, Shanghai, Zhejiang, Fujian, and Guangdong, boasts the highest offshore wind-generation potential within China's exclusive economic zone (EEZ). The total installed capacity and electricity generation from wind power in East and South China account for approximately 20% of the national total, primarily driven by offshore wind. It is worth noting that the eastern coastal provinces use approximately 50% of their total potential capacity, with Shanghai at 55%, Zhejiang at 55%, and Fujian at 49%.

Solar PV installations are gradually expanding outward from urban load centers (see definition in the ESI,[†] Section S3.3.2) due to land availability constraints and an overall lower capacity factor, resulting in greater sensitivity to integration distances (spur and trunk lines, see the ESI,[†] Fig. S69). From 2030 to 2060, 80% of the cumulative installed capacity extends from within 60 km of load centers to within approximately 75 km (see the ESI,[†] Fig. S70). The primary regions for solar PV deployment are North, East, and South China, which together account for 60% of the nation's installed capacity and electricity generation. By 2060, stricter emissions constraints will lead provinces with limited suitable land and high electricity demand to nearly deplete their solar PV potential. For instance, solar PV installations will exhaust over 80% of the potential capacity in Shanghai (99%), Guangdong (92%), Zhejiang (89%), Beijing (88%), Tianjin (84%), and Jiangsu (83%), as illustrated in Fig. 4E. After 2050, significant growth occurs in large-scale solar PV bases in Mengxi and Northwest China, as shown in Fig. 4B, with more concentrated installations and higher installed capacity per grid cell. This expansion leverages the abundant unused land in these regions suitable for utility-scale solar PV development. Xizang, which possesses the richest solar irradiance across China but is far from demand centers, sees only 113 GW of installed capacity and 218 TW h per year of electricity generation from solar PV installation in 2060.

We developed an XGBoost model to estimate the rooftop area potential^{46,47} and adopt the optimal fixed-tilt photovoltaic model¹⁶ to assess the installation capacity potential for distributed solar PV (DPV) systems in each grid cell (see the ESI,[†] Section S2.1.5 for details). In the CISPO model, distributed solar PV systems are analyzed assuming zero transmission costs for local integration and higher investment costs than utility-scale solar PV (UPV) systems for rooftop rentals.⁴ By 2030, DPV installations are projected to reach 903 GW, an increase of approximately 750 GW from 2022, generating 1.15 PW h per year, and UPV is expected to reach 885 GW, producing 1.27 PW h per year. These installations will occupy more than 10 000 km² of rooftop area for DPV and 11 000 km² of land area for UPV, approximately 3–4 times the land area of Shanghai city (about 6300 km²), as shown in Fig. 5. Post-2030, DPV installations will have a mild increase, reaching 930 GW by 2040 and 1008 GW by 2050. However, by 2060, the capacity of DPV is projected to decline significantly to 489 GW, reducing the occupied rooftop area by 50% to 5387 km² (see the ESI,[†] Fig. S63, for the spatial distribution of DPV from 2030 to 2060).

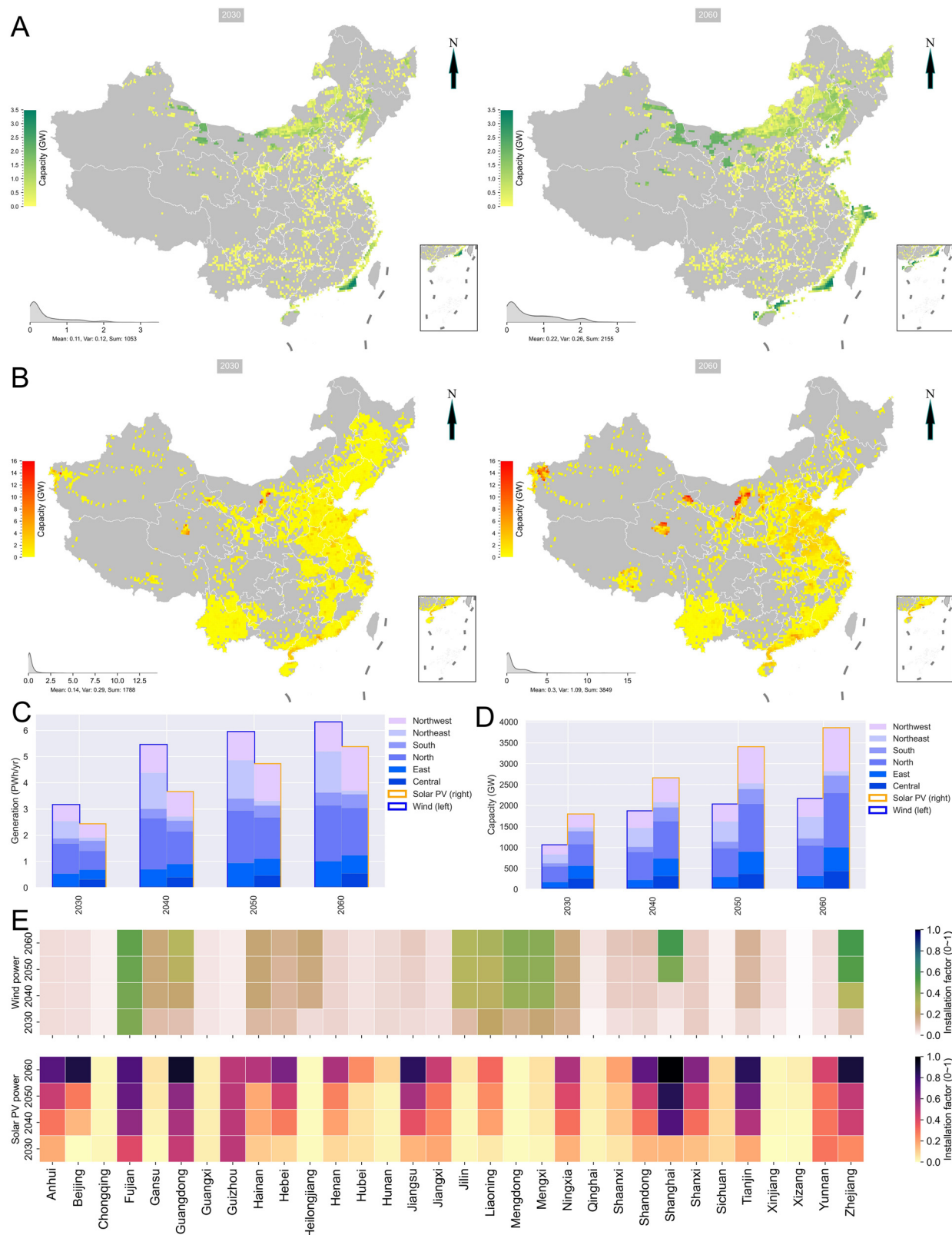


Fig. 4 Optimized deployment of VRE to 2060. (A and B) Capacity (GW, new + existing) of wind (onshore + offshore) and solar PV (utility-scale + distributed) in each cell. (C and D) Annual power generation (PW h per year) and installed capacity (GW) of wind (left) and solar PV (right) in each grid region. (E) Provincial installation factor (0–1), defined as the ratio of planned capacity to the installation capacity potential, from 2030 to 2060.

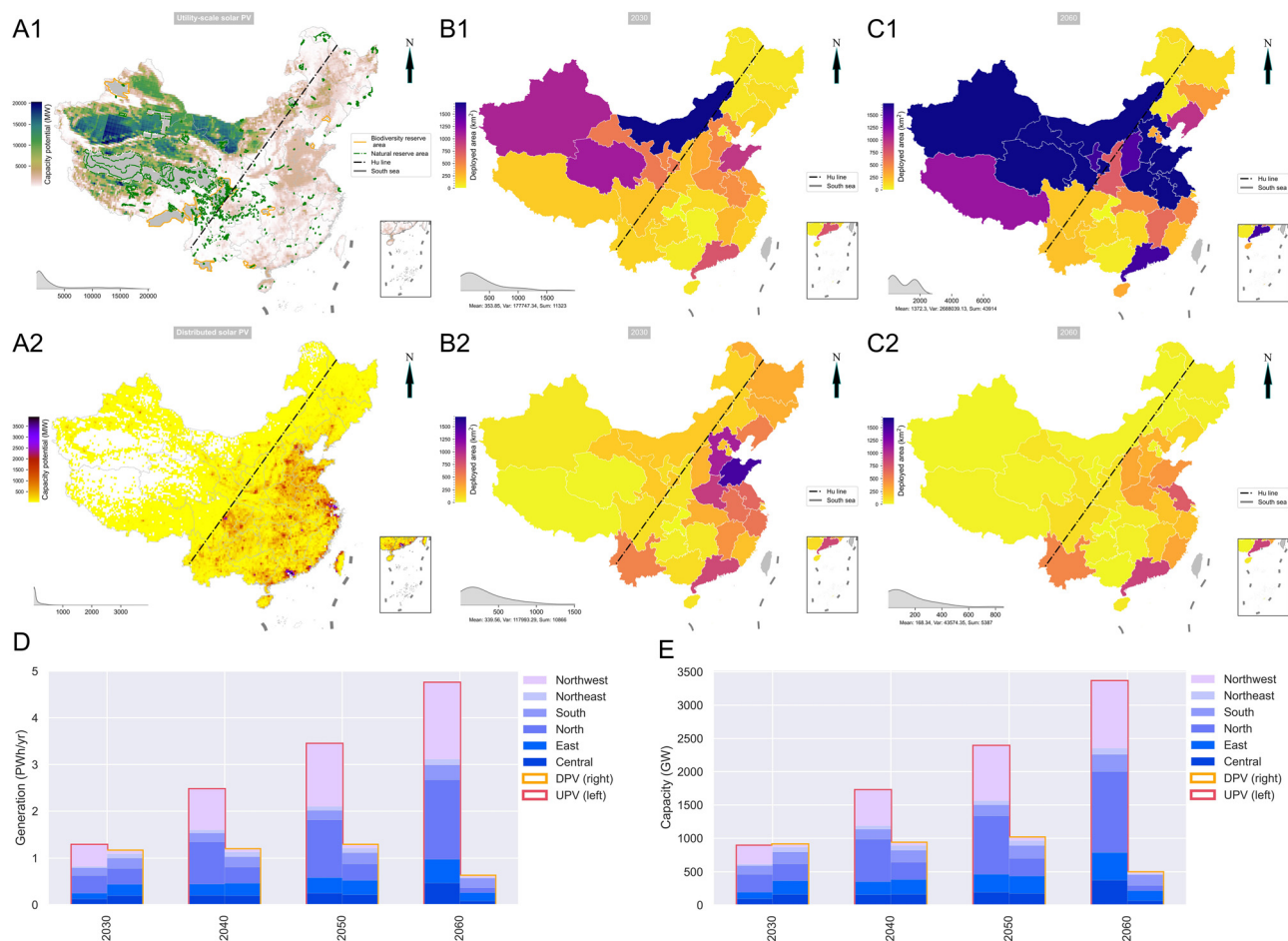


Fig. 5 Installation capacity potential and optimized results for utility-scale and distributed solar PV systems. Cell-level installation capacity (MW) potential (A1 and A2), provincial installation area (km²) in 2030 (B1 and B2), and 2060 (C1 and C2) for UPV (upper) and DPV (lower) systems. Annual generation (D) and installed capacity (E) for UPV (left) and DPV (right) systems from 2030 to 2060.

This notable decrease is primarily attributed to the retirement of DPV installations in 2030, the adoption of other renewable energy sources (e.g., offshore wind in Jiangsu, Shanghai, and Zhejiang), and the long-distance transmission delivering power from the west and north to the east and south, which are more competitive under our cost assumptions in 2060. Instead, UPV installations are expected to quadruple compared to the 2030 level, reaching 3359 GW and occupying 43 914 km² of land area by 2060.

Land-use tension between energy generation and other sectors (e.g., agriculture) is a critical challenge for the successful deployment of TW-scale VRE. This study evaluates the installation capacity potential of wind and solar PV systems across three land-use scenarios: open, base, and conservative. These scenarios reflect varying levels of land-use restrictions, with suitability factors for each land-use type assigned to prioritize government objectives. A detailed assessment of VRE potential is provided in the ESI,[†] Section S2.1. Under the base land-use scenario, the estimated installation capacity potentials are approximately 14 000 GW for onshore wind, 1500 GW for offshore wind, 46 000 GW for utility-scale solar

PV, and 1500 GW for distributed solar PV. Expanding suitable land availability under the open scenario increases potential by 21%, 74%, 41%, and 6.7% for onshore wind, offshore wind, utility-scale solar PV, and distributed solar PV, respectively, reaching 17 000 GW, 2700 GW, 65 000 GW, and 1600 GW. Conversely, adopting more conservative assumptions reduces potential by 39%, 33%, 41%, and 16.7%, respectively, yielding 8600 GW, 1000 GW, 27 000 GW, and 1250 GW.

System impacts of the available land area are assessed by incorporating the installation capacity potential under different land-use scenarios (“VRELandOpen” and “VRELandConservative”) as upper deployment limits for each VRE grid cell within model constraint S4-2 (see ESI,[†] Section S4.3.1). From 2030 to 2060, power system costs decrease (increase) by 0.1–2.5% (1.6–3.3%) under the open (conservative) land-use assumptions for VRE, as shown in Fig. 2. At the national level, the total installed capacity of wind and solar power (including CSP) exhibits minor variations across the land-use scenarios. For instance, in 2060, the base scenario features 6132 GW of installed capacity, while the “VRELandOpen” and “VRELand-Conservative” scenarios result in 5768 GW and 5780 GW of

installed capacities, respectively. Under more restricted land-use assumptions, distributed solar PV and CSP installed capacities increase twofold from the base scenario, reaching 1044 GW and 322 GW in 2060, respectively, highlighting the potential of these technologies to mitigate land-use pressures.

2.4 Supporting role of energy storage and inter-regional transmission

Fig. 6 shows the hourly energy stored in PHS and BAT storage systems over 8760 hours at the national level, representing the aggregated values across all regions. In this figure, the horizontal axis represents each day of the year, and the vertical axis denotes the hours from 0 to 23 for each day. The color shades indicate the energy stored at the national level, with darker shades representing higher values. Generally, these energy storage systems' charging/discharging patterns exhibit a strong correlation with the temporal characteristics of solar PV generation. Specifically, electricity is stored during peak PV generation periods (typically 12:00–18:00 with gradually deepening color) and discharged primarily during night hours (as the color gradually lightens) (see Fig. 6 and the national power dispatch profiles in the ESI,[†] Fig. S81). PHS and BAT exhibit

distinct operational patterns as the proportion of integrated VRE varies. PHS exhibits a more significant role in long-term regulation in 2030 compared to 2060, as evidenced by sustained multiple-day charging periods (color intensifies and remains) followed by multiple-day discharging periods (color gradually fades), for example, from days 210 to 230 in Fig. 6. This difference is attributable to the higher VRE penetration in 2060, which increases the frequency of energy storage cycles. BATs, because of their shorter discharge duration (typically 4 hours), primarily serve a rapid balancing function. However, when the installed capacity exceeds 2000 GW h in 2060, BAT storage can also provide longer-span balancing to some extent, as depicted in Fig. 6 between days 270 and 300.

Inter-regional ultra-high voltage (UHV) transmission lines are essential for addressing the resource–demand mismatch in China.^{2–4} The CISPO model optimizes the construction and reinforcement of UHV lines with voltages larger than or equal to 500 kV between any pair of provinces, as detailed in the ESI,[†] Section S3.7. Our findings indicate that by 2060, the nationwide capacity of inter-regional transmission lines needs to triple from the 2022 level, reaching approximately 2.0 TW. Fig. 7 shows the annual inter-provincial electricity flows (TW h per year) along these

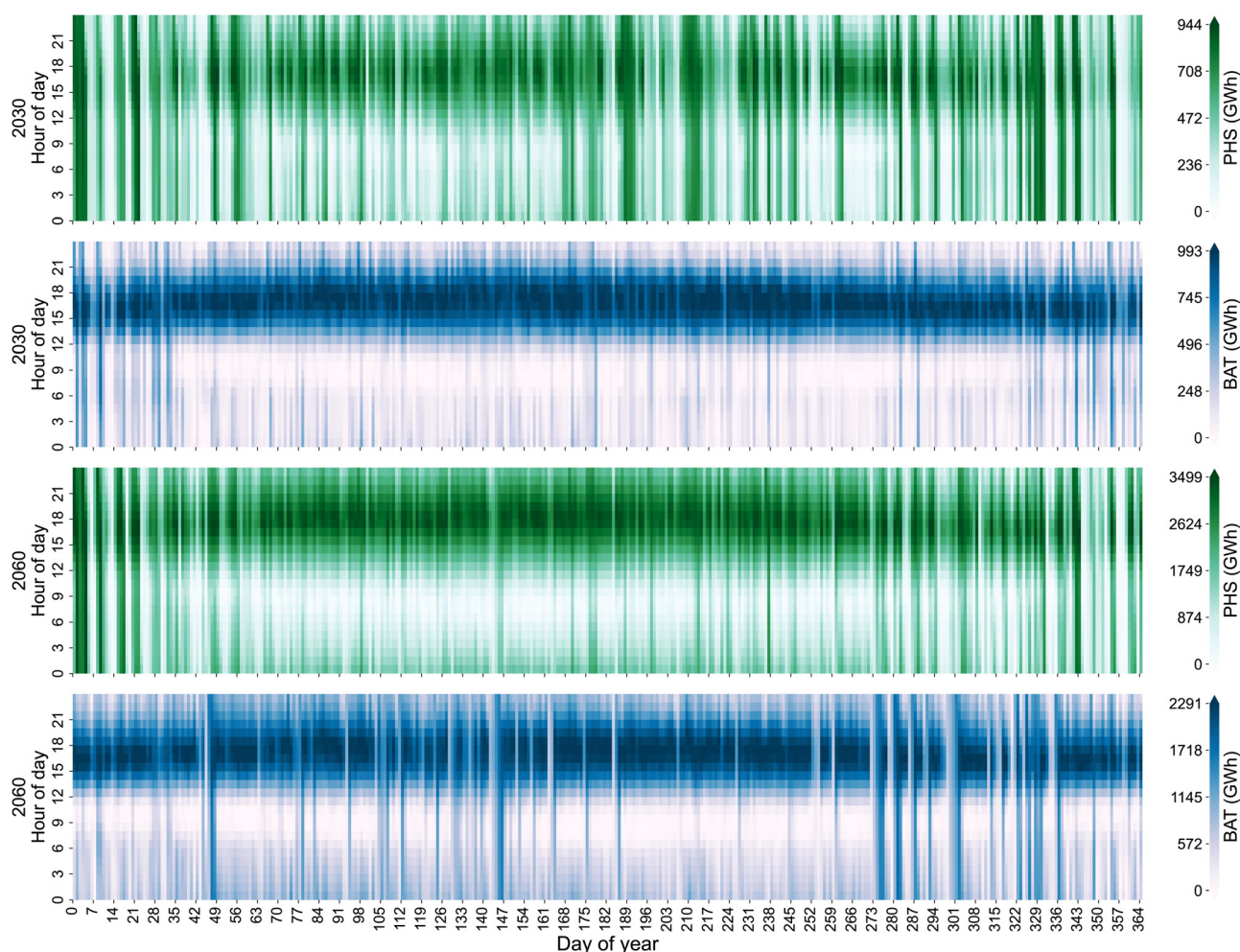


Fig. 6 Results of the total energy stored (GW h) in PHS and BAT storage systems at the national level for each hour in 2030 and 2060.

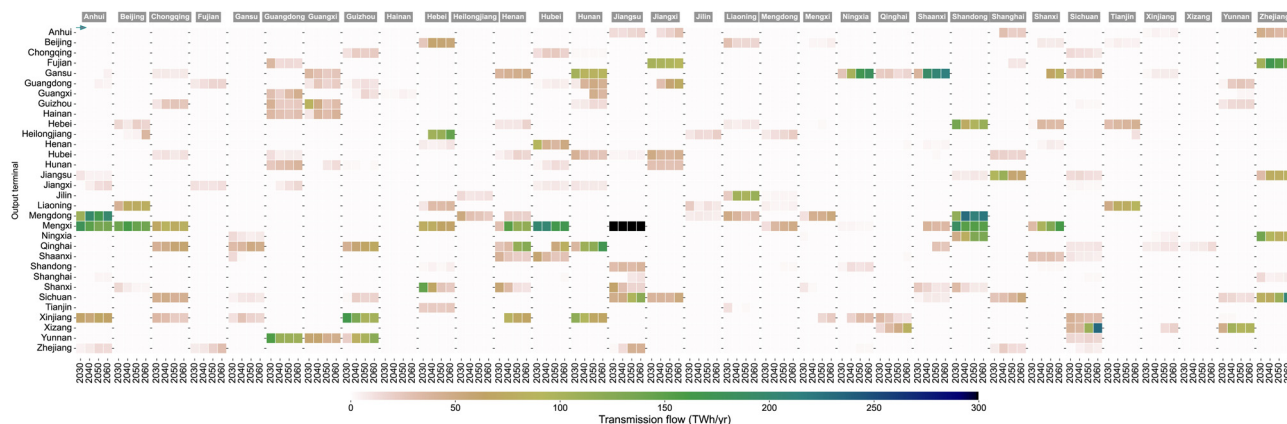


Fig. 7 Results of yearly energy transmitted (TW h per year) through inter-regional transmission from 2030 to 2060. The left axis is the power output province.

transmission lines, with the left vertical axis representing the exporting end. The results demonstrate that the inter-regional transmission lines serve two primary functions. First, they enable the transfer of wind and solar power from resource-rich regions to areas with high electricity demand, such as the transmission of multi-petawatt hours of electricity from Mengxi and Mengdong to provinces like Jiangsu and Anhui, while having minimal reverse flows (see the net transmission flow (TW h per year) maps in the ESI,[†] Fig. S68). Second, they play a balancing role in regions with high electricity demand, such as East, Central, and South China, as evidenced by bidirectional power flows along lines like Anhui–Jiangsu, Anhui–Jiangxi, Chongqing–Hubei, and Guangdong–Yunnan.

2.5 System volatility and capacity payment mechanism

In a competitive setting, the market clearing price would equal the marginal cost (MC) of generation, when plants earn returns for supplying electricity.⁴⁸ Under the linear optimization scheme, the hourly MCs of generation for each province are derived from the shadow prices of the power demand balance constraints. With the increasing integration of VRE sources from 2030 to 2060, the system MCs exhibit an overall trend of decreasing prices accompanied by heightened variability, owing to the zero marginal cost and intermittent nature of VRE generation, as shown in Fig. 8. In 2030, most MCs fall in the range of 0.2–0.4 yuan per kW h, with a small proportion in a comparably lower interval from 0.0 to 0.1 yuan per kW h. By 2040, a greater MC share is found in the range of 0.0–0.1 yuan per kW h, while the highest MC reaches approximately 1.0 yuan per kW h. Beyond 2050, we see an increasing number of low MCs and instances of extremely high MCs, ranging from several to tens of yuan per kW h. Statistically, the mean and standard deviation of the hourly MCs in 2030 are 0.28 and 0.09, respectively, shifting to 0.31 and 0.6 by 2060. These results indicate that the power system volatility becomes increasingly pronounced for large-scale VRE deployment (see the hourly MC distribution of other scenarios in the ESI,[†] Fig. S71–S80).

Firm resources (*e.g.*, thermal and nuclear power) are the key electricity generation systems that support a more stable power

system when large-scale weather-dependent energy is integrated. Our results show that the coal power plant is the main supplemental capacity to enhance the system's stability when the MCs are high, indicating inadequate capacity supply in the power system. For example, we observe peak MC periods for insufficient renewable energy supply (particularly in winter when hydro discharge is limited) and peak power demand (in summer) in 2050 and 2060 (Fig. 8), at which point coal power fills the capacity gaps (see the national hourly power supply-demand profile in the ESI,[†] Fig. S81).

China implemented a capacity payment mechanism to compensate coal power plants⁴⁹ in 2022, as a response to the decreasing revenue from the electricity market and increasing expenditure of more frequent shut-downs, start-ups, and ramping. In this study, the net profit of power generators is determined by first computing annual revenue as the cumulative sum of hourly electricity generation (kW h) multiplied by the system marginal generation cost (yuan per kW h), then subtracting annualized capital expenditures and operational expenditures. When the net profit is negative, a capacity payment is required to maintain market viability. In 2030, 25 out of 32 provinces need the capacity payment for coal plants (161–426 yuan per kW year), with most provinces exhibiting values between 300 and 400 yuan per kW year (see Fig. 9A). From 2030 to 2050, most provinces will see a slight decrease in coal capacity prices compared to those in 2030, while several provinces experience increases, *e.g.*, Yunnan and Hebei.

Similar to the firm generators, energy storage systems (ESSs) can also help balance the electricity supply-demand in a high-penetration VRE power system³⁸ and provide essential services, such as inertia support³ and spinning reserves.^{2–4,28} Our results show that while approximately 5800 GW h of energy capacity is recommended to be deployed in 2060, 80% of the ESSs have negative profits if there is only revenue from arbitrage in the electricity market. In 2030, among the nearly 2000 GW h of ESS capacity, PHS is projected to incur losses ranging from 250 to 400 yuan per kW year in most provinces. Battery storage losses in 2030 are expected to range between 200 and 300 yuan per kW year. From 2040 onwards, the losses associated with ESSs are

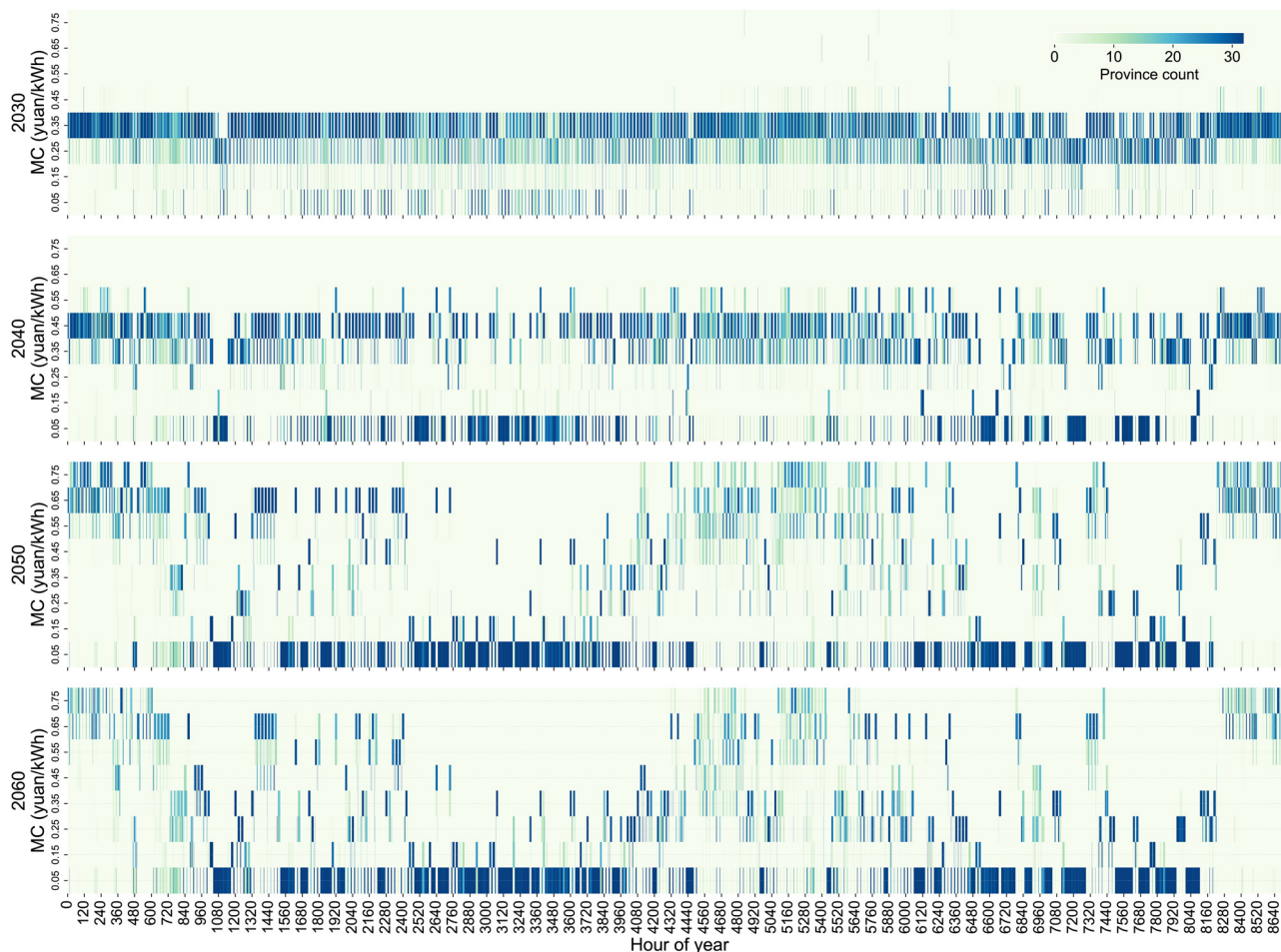


Fig. 8 Results of the hourly provincial marginal cost of generation (MC, yuan per kW h) from 2030 to 2060. We divide the MC into eight intervals (0.0–0.1, 0.1–0.2, ..., 0.7–0.8), and count the number of provinces located in each interval for 8760 hours. MCs higher than 0.8 yuan per kW h are also counted in the 0.7–0.8 interval.

found to narrow, primarily benefiting from the larger MC ranges and arbitrage opportunities. PHS is expected to exhibit superior profitability, with losses decreasing to around 120 yuan per kW year (about –52 to –70%) in 2050 and 50 yuan per kW year (about –80 to –87.5%) in 2060 in most provinces. By 2060, we observe that pumped hydro storage may earn positive profits in provinces such as Ningxia and Qinghai (see Fig. 9B). For battery storage, losses could decrease from around 150 yuan per kW year in 2040 to approximately 50 yuan per kW year in 2060, but positive profitability remains elusive. This finding hints that energy storage technologies with lower energy costs and higher duration hours might be relatively more competitive in the future power system landscape.

2.6 Sensitivity of cost and electricity demand projection

We investigate the impact of slower investment cost reductions for key technologies – wind and solar power, PHS, BAT, and BECCS – on the transition of China's power system. In the “Capex1.25X” scenario, the capital expenditures and associated fixed operation and maintenance (O&M) costs of these technologies are increased by 25% compared to the base scenario,

reflecting potential variations in technological advancements across key energy sectors. System costs increase by around 10% (approximately 500 billion yuan per year) in 2060 under these higher cost assumptions (Fig. 2). In the base scenario, annual electricity demand is projected to rise from 12.5 PW h per year in 2030 to 16.0 PW h per year in 2060, aligned with forecasts from economy-wide models^{6,35} and expert assessments.^{13,14} To assess the sensitivity of higher electricity demand, we introduce the “Demand1.25X” scenario, where annual demand is 1.25 times that of the base scenario, increasing from 15.6 PW h per year in 2030 to 20.0 PW h per year in 2060, accommodating potential growth in carbon-free electricity applications such as power-to-gas. Under these demand assumptions, the total installed capacity of wind and solar PV energy expands significantly, reaching 7853 GW (+31% compared to the base scenario) in 2060. Concentrating solar power causes a threefold increase compared to the base scenario, with the installed capacity reaching 380 GW and annual generation reaching 1.0 PW h per year by 2060. Despite the substantial increase in annual demand, the system cost of electricity remains comparable to the base scenario (Fig. 2).

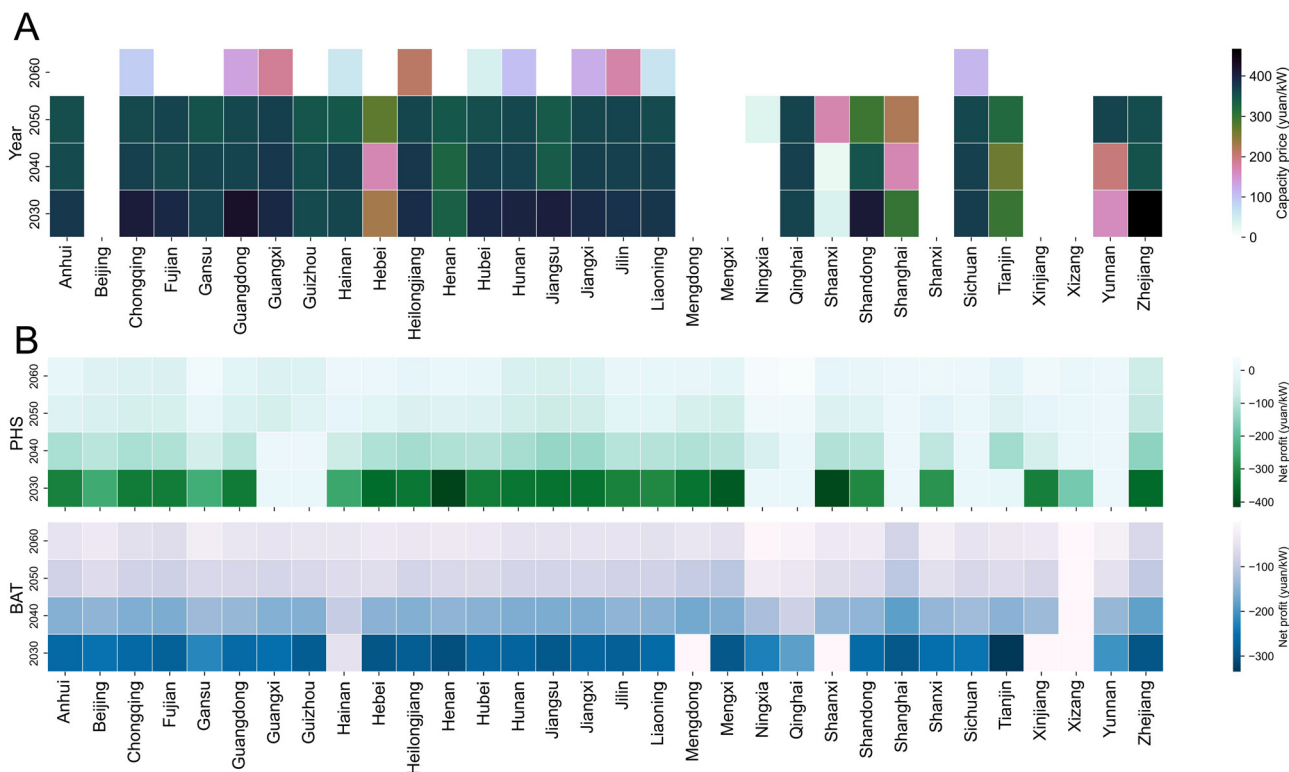


Fig. 9 Results of capacity payment (yuan per kW year) to coal power (A), and net profit (yuan per kW year) for PHS and BAT storage (B) from 2030 to 2060.

2.7 Sensitivity of PSEM to spatiotemporal resolution

The increasing penetration of variable renewable energy in power systems requires higher spatial resolution and broader temporal coverage for renewable energy and energy storage deployment decisions in PSEMs.^{4,22,23,25} To illustrate this fact, we introduce variants of the CISPO model by employing traditional modeling approaches to examine the impacts of spatial and temporal resolution in 2060, aggregating renewable resource profiles into a limited number of clusters per load zone or optimizing across representative time slices.^{2,3,50–52} The objective is to demonstrate biases arising from simulation granularity. First, the “VREcluster” scenario integrates each $0.25^\circ \times 0.25^\circ$ grid cell within provinces into resource clusters stratified by VRE types, hourly capacity factors, and proximity to load centers (detailed methodology available in the ESI,[†] Section S3.3.3). Second, for different temporal coverages, we designed a total of 22 distinct temporal coverage scenarios, with the number of hours varying from 288 to 4320. The different “period” intervals are designed to account for representativeness across months or seasons. Scenario design and temporal coverage details are provided in the ESI,[†] Section S6 and Fig. S47. Significant biases emerge for VRE installations in scenarios with lower spatiotemporal modeling resolution. Capacity deviations range from -28.0% to $+8.8\%$ for onshore wind, -36.8% to $+46.6\%$ for offshore wind, -8% to $+17\%$ for utility-scale solar PV, and -64% to $+76\%$ for distributed solar PV in 2060 (Fig. 10A and B). Compared to cell-level optimization, the clustered VRE scenario concentrates development in

areas with higher resource potential, while some are very far from load centers (see the ESI,[†] Fig. S65). By omitting the integration costs within a cluster, the “VREcluster” scenario underestimates system costs by 9% in 2060.

For scenarios with temporal coverage limitations, the optimal deployment of energy storage systems, which are highly sensitive to chronological strategies for balancing weather-dependent renewable energies, is particularly biased. Discontinuous temporal representations – characterized by skipping months or seasons in representative hours – disrupt the continuity of actual weather conditions, and consequently misestimate the role of energy storage technologies with varying temporal durations. For instance, solar PV output exhibits substantial variability between seasons, such as spring and summer,^{30,53} so models may overestimate battery capacity needs for balancing solar PV output when inappropriately concatenating temporally distant typical days. The analysis of different temporal coverage scenarios reveals significant overestimation in battery storage deployment at the national level: installed capacities are higher in all of these scenarios, with the largest overestimate up to $+50\%$ in 2060, compared to the base scenario.

3 Discussion

In this study, we developed a temporally and spatially resolved optimization model to explore the transition pathway of China’s power system toward negative carbon emissions. Our

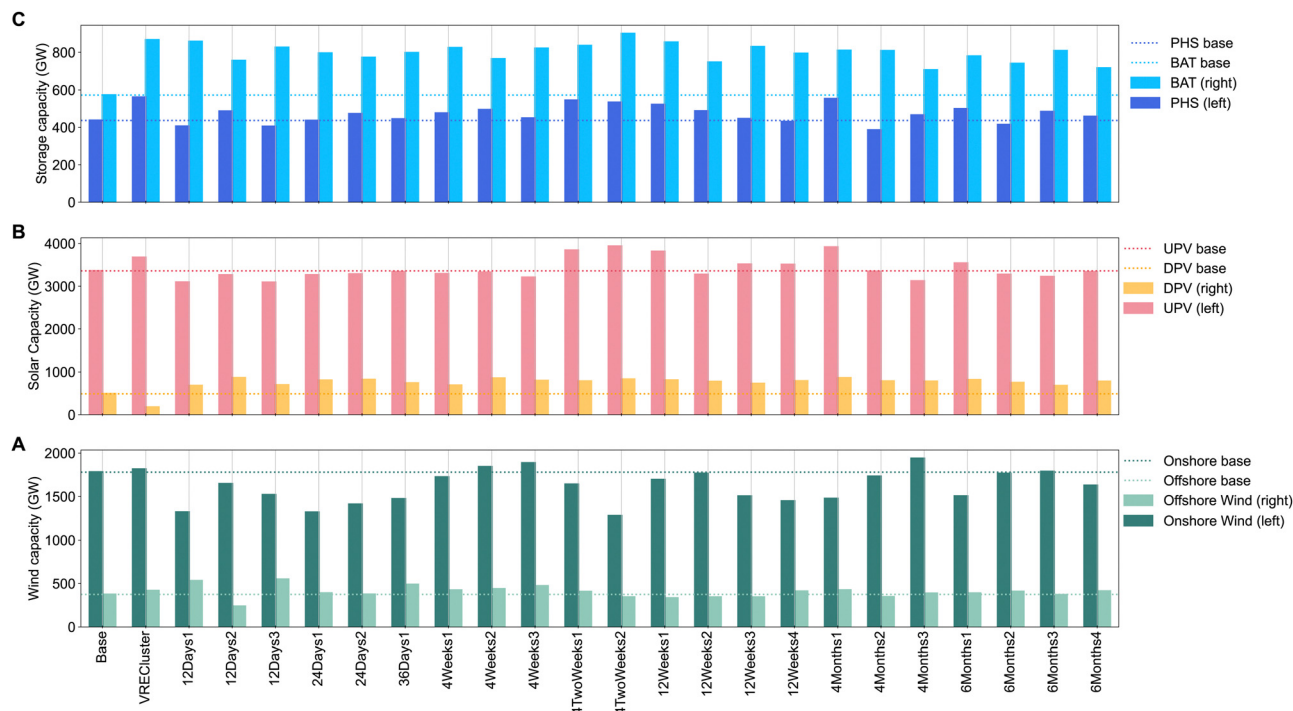


Fig. 10 Sensitivity analysis of the PSEM for results in 2060 under 23 different spatiotemporal resolution scenarios. (A) Capacity (GW) of onshore wind (left) and offshore wind (right). (B) Capacity (GW) of utility-scale (left) and distributed (right) solar PV. (C) Capacity (GW) of pumped hydro storage (PHS, left) and battery storage (BAT, right).

results show strategies of the electricity sector to attain negative 200–700 MtCO₂ per year by 2060, which entails a large-scale deployment of approximately 6000 GW of wind and solar power, 5800 GW h of energy storage, 2000 GW of inter-regional transmission, and 850 MtCO₂ per year of carbon capture and sequestration. These findings have several policy implications for both near- and long-term planning. First, distributed solar PV power generation is projected to experience substantial installation growth, reaching 930 GW by 2030. However, an effective integration guarantee or power compensation mechanism needs to be designed and implemented to encourage integrating generation from distributed solar PV power generation because of the high penetration of VRE presenting significant challenges for the grid.^{54,55}

Second, our results indicate that the pathway of China's power system toward negative emissions can be divided into two distinct phases. The first phase, spanning from the present (2022) to around 2045, is characterized by sustained and large-scale deployment of carbon-free energy sources to substitute fossil fuels. The second phase, from 2045 to 2060, necessitates the implementation of BECCS or DACCS at a scale of 100–200 GW or 500–1000 MtCO₂ per year to attain net-zero and ultimately negative emissions. Although BECCS is more cost-effective than DACCS as a NET in China's power system, the potential barriers, such as land-use availability, water resource constraints, and public acceptance,^{42,56} may limit the deployment of BECCS. For this consideration, DACCS is another possible option for negative emissions in areas where deploying BECCS is not feasible. Additionally, while the model suggests

that deploying CCS technology in traditional thermal power plants is not immediately necessary, the large-scale deployment of BECCS or DACCS requires technology maturity and cost reduction in the related carbon capture and sequestration industry. Therefore, it is essential to formulate planning for piloting carbon capture in thermal power units, transportation, and sequestration to foster technological advancement and readiness.

Third, our results demonstrate that carbon prices and complementary policy instruments would be crucial to promote the low-carbon transition of the power system. China's national emissions trading scheme (ETS) was launched in 2021 with a current carbon price level of around 100 yuan per tCO₂. With a decreasing supply of allowances and increasing share of allowances being auctioned, the ETS will introduce a growing carbon price,⁵⁷ which is expected to increase the cost of fossil fuel generation and enhance the competitiveness of renewable energy, abated fossil fuel generation (coal or gas power plants with CCS), and ultimately, BECCS in the wholesale power market. At the same time, our modeling analyses point out that revenue from the wholesale market cannot always ensure the cost recovery for (abated) coal power plants and battery storage, prompting the need for introducing capacity mechanisms or subsidies to battery storage. Therefore, policymakers should also be aware of these important interactions and coordinate the design of complementary policy instruments effectively. Modeling tools like the CISPO model will help policymakers in China and beyond analyze these interactions quantitatively.

This study has several limitations, though we believe they do not alter the main findings. First, technologies not explicitly modeled include geothermal power, floating offshore solar PV, and offshore carbon sequestration. Second, the model employed here aims to simulate long chronological (8760 hours) variability from renewable energy sources and power demand at the hourly resolution. Non-linearities arising from unit commitment, optimal power flow, and other factors are modeled using simplified approaches that have been validated with negligible loss of fidelity in previous work.^{2,3,23,29,51,58,59} Third, analyses of additional environmental impacts, such as land-use changes and water consumption, as well as other important socioeconomic factors, would enable a more comprehensive perspective on the multifaceted challenges and trade-offs inherent in the transition to a net-zero society.

In summary, the CISPO model developed in this study provides detailed representations of power system expansion and operations, offering valuable insights into pathways for decarbonizing China's power sector to mitigate global climate change. The integrated modeling approach, combining high spatial and temporal resolution with a diverse portfolio of technologies, is highly adaptable and can be readily applied to other regions exploring possible strategies to transition from carbon-intensive²⁸ to sustainable energy systems. First, the imbalance between renewable energy resource availability and power demand is a global challenge,²¹ observed in regions such as the United States, India, Europe, and South America. This underscores the critical need for integrated transition analyses that encompass variable renewable energy, hydropower, biomass energy, conventional thermal power, negative emissions technologies, energy storage, and long-distance transmission. Second, the comprehensive assessment of energy-related resources leverages widely utilized open-source datasets, including elevation, slope, water discharge, and reanalysis data of meteorological factors, providing global coverage. The methodologies employed in this study can be easily expanded to other countries using these readily available datasets. Third, the high spatiotemporal resolution power system model incorporates a flexible framework that can be readily adapted to other regions or even the global scale. This adaptation can be achieved through the use of localized parameters, such as investment costs, existing power infrastructure, and region-specific power demand profiles. Looking ahead, the model can be expanded to incorporate analyses of geospatial constraints, such as trade-offs between land use and renewable energy deployment, impacts of long-chronological renewable energy supply uncertainties (e.g., extreme weather) on power system operation, and involving global renewable energy supply chain and critical materials challenges.

4 Materials and methods

4.1 Energy resource assessment

We assess the resources of wind (onshore and offshore) and solar (utility-scale PV, distributed PV, and CSP) power by hourly

capacity factor (CF, 0–1) and installation capacity potential (MW). The meteorological data including wind speed (10 m, and 100 m), surface short-wave radiation, temperature, and so on, are derived from the European Centre for Medium-Range Weather Forecasts Reanalysis Version 5 (ERA5) dataset, which provides a record of global climate and weather spanning over eight decades with a spatial resolution of $0.25^\circ \times 0.25^\circ$ and hourly temporal resolution.⁶⁰ With these high-resolution meteorological data, we adopt the wind turbine model parameters from the National Renewable Energy Laboratory (NREL)⁶¹ and the generation model widely used^{20,23,62–64} to calculate the hourly CF of wind and solar power in each grid cell, respectively. The maximum installation capacity potential of onshore wind, offshore wind, utility-scale solar PV, and CSP systems is estimated first with the suitable area constrained by natural/biodiversity reserves, technical conditions, and land-use type.⁴ We collect as much vectorized rooftop data as possible from around the world,⁶⁵ which is recognized by satellite images and computer vision algorithms, and then train an XGBoost regression model using population, road length, night light, and built-up area as independent variables, and rooftop area as dependent variables, in cells with fine rooftop data to predict those cells without reliable vectorized rooftop data.^{46,47} The suitable area in each cell to deploy distributed solar PV power is determined as a fraction of the rooftop area. Within the suitable area (km^2) in each cell, the installation capacity potential is derived by multiplying the installation density (MW km^{-2}) with assumptions for wind power using 7×7 rotor diameters spacing,^{29,66} and for solar PV power using the fixed-tilt model.¹⁶ See the ESI,[†] Section S2.1, for the detailed methodology, dataset, and results.

Other energy-related resources are also assessed in this study, including hydropower installation and output potential, available biofuel resources, and carbon sequestration limitations. We assess the installation capacity potential of hydropower in China through an analysis of the least levelized cost of energy (LCOE), taking into account factors including the inundated area, environmental conservation, population displacement, and integration distance, as documented in previous studies.^{67,68} Under this assessment framework, the optimal dam height (m), reservoir capacity (m^3), and installation capacity potential (MW) are determined at the least LCOE, which are key inputs to model hydropower deployment. With this installation capacity potential, we use the discharge data ($\text{m}^3 \text{s}^{-1}$) of a 3-hour temporal resolution in 2019⁶⁹ to simulate the generation of run-of-river and reservoir hydropower, in which the hourly discharge is assumed to be the same in each 3-hour interval. See the ESI,[†] Section S2.2, for details on hydropower potential assessment. We use the agricultural residues, forest residues, and energy crops on abandoned cropland with quantified potential in TJ per year for each $1 \text{ km} \times 1 \text{ km}$ grid cell from Wang *et al.*⁴³ to aggregate these potentials at the provincial level and then estimate the installation capacity potential using assumptions for thermal efficiency at 0.35⁷⁰ and annual equivalent hours at 6132 hours. The installation capacity potential of biomass under this estimate is about 470 GW

(see the ESI,[†] Section S2.3, for details on biofuel resource distribution). In this study, we primarily focus on onshore deep saline aquifers (DSA) as a preferred option for carbon storage due to their substantial capacity within China.²⁶ The World Geologic Provinces dataset with further adjustment based on the findings of Wei *et al.*²⁶ and Wang *et al.*¹⁵ is used to estimate the surface area encompassing all sedimentary basins. Nationally, we find approximately 120 Gt CO₂ storage potential of onshore deep saline aquifers. See the ESI,[†] Section S2.4, for the estimate formula and results.

4.2 Integrated sustainable power system optimization model

The CISPO model developed here minimizes the sum of the annualized capital cost of investments and operational costs over a full year (8760 hours) of hourly operation using 2019 meteorological data, subject to meeting hourly electricity demand and a variety of engineering, economic, and policy-related constraints in a planning year. Technologies optimized in the CISPO model include wind (onshore and offshore), solar (utility-scale PV, distributed PV, and CSP), hydropower (run-of-river and reservoir), firm generator (coal, gas, nuclear, and bioenergy), energy storage (BAT and PHS), CCS, direct air capture, and transmission (intra- and inter-region). In each planning year, the CISPO model outputs these technologies' optimized deployment and hourly operational status. The spatial installation resolutions for VRE, hydropower, and the remaining generators are grid cell ($0.25^\circ \times 0.25^\circ$), dam site, and province, respectively. The information transferred between different planning years (*e.g.*, from 2050 to 2060) covers newly built and retired generation capacities. We model the hourly operation considering available VRE output, reservoir capacity and output of hydropower, unit-commitment of thermal and nuclear power, charge/discharge profiles of energy storage, spinning reserve, inertia requirement, and power transmission flow through the UHV lines. Within the high spatial and temporal resolution for capacity expansion and power system operation, the CISPO's scale comes to about 4×10^7 rows, 3×10^7 columns, and 5×10^8 non-zero elements after pre-solve. See model comparisons of the CISPO with previous PSEMs in the ESI,[†] Section S1.

The grid integration distance and capacity of renewable energies include spur and trunk lines, which are associated with the decision variables of deployment following the methodology used in ref. 4 and 23. We collect the location information of substations with voltage levels larger than 220 kV from Open Street Map (OSM)⁷¹ as the spur line terminates. Major load centers are recognized as urban areas under 1:50 M zooming recognition level,⁷² and the additional load center in Xizang is added from the 1:10 M level to guarantee that there is at least one major node designated in Xizang. Within the location information of renewable energies, substations, and load centers, we adopt a minimize-integration-distance algorithm to determine the connecting pair. See the ESI,[†] Section S3.3.2, for detailed results. We endogenously model the optimization matching of carbon capture and sequestration and allow each province to transport the captured CO₂ to any

sequestration sites, with the transportation distance estimated using the geographical distance between province centers and sequestration locations. See the ESI,[†] Section S3.8, for the unit capture, transportation, and injection costs. The relaxed unit commitment algorithm for thermal and nuclear dispatch is adopted in the CISPO model as previous studies have proved this approach runs with much higher solving speed and negligible loss ($<1\%$).^{2,58} Similarly, we use the transportation (or pipeline) model to optimize the operation of inter-regional transmission which is validated⁵⁹ and used in many other PSEMs.^{2–4,23,28,51,52} Detailed model formulas are shown in the ESI,[†] Section S4.

4.3 Input data and assumptions

The CISPO model covers 32 provincial load areas of mainland China, with Inner Mongolia divided into west (Mengxi) and east (Mengdong) regions because they belong to different power grids. These load areas are modeled for demand balance, inter-regional transmission flow, and policy requirements. See the ESI,[†] Table S1 and Fig. S2, for detailed region divisions.

In this study, we take 2022 as the base year and optimize China's power system from 2030 to 2060 (10 years as a step). The installed capacities of coal, gas, nuclear, wind, and solar power by 2022 are derived from the Global Energy Monitor.⁷³ This dataset reports the unit-level information of installed capacity, including operation status (operational or under construction), remaining lifetime, and locations. By 2022, we find that the total operational capacity is 1108 GW for coal power (including combined heat and power plant), 108 GW for gas power, 56 GW for nuclear power, 278 GW for onshore wind power, 30 GW for offshore wind power, and 228 GW for utility-scale solar PV power. We have collected a total of approximately 360 GW of installed hydropower (run-of-river and reservoir) at the dam site level from Wan *et al.*,⁷⁴ 46 GW of installed pumped hydro storage capacity from the National Energy Administration (NEA),³⁹ 156 GW of installed distributed solar PV power from NEA,⁷⁵ and 41 GW of installed biomass power from China Electric Power Planning & Engineering Institute.⁷⁶ Under-construction coal, gas, nuclear, and PHS projects are assumed to be in operation by the year 2030. In each planning year (*i.e.*, 2030, 2040, 2050, and 2060), the installed capacity of coal and gas power will be retired if reaching the designed lifetime, or retrofitted to be equipped with carbon capture equipment if needed. We assume that the installed wind power, solar power, hydropower (including PHS), and biomass power will be renewed if retired. See the ESI,[†] Section S3.3, for details of the generator installations, and the capacity allocation to each grid cell for VRE.

To generate the 8760-hour electricity demand profiles of the 32 provinces, we first collect data on the typical hourly demand loads for weekdays and non-workdays, as well as the daily maximum and minimum loads observed throughout 2019 as detailed in ref. 77. Subsequently, we adjust the hourly demand load for each day based on whether it falls on a weekday or non-workday, and the corresponding maximum and minimum demand levels. The resulting hourly demand for each day is

then concatenated sequentially to construct the annual 8760-hour profile. Following this, we scale up the hourly demand profile for the base year (2022) using the annual demand increase rates relative to 2019 in each province, as outlined in the ESI,[†] Section S3.1, with Fig. S25–S28 illustrating the normalized load profile. Provincial 8760-hour demand profiles are further scaled using the annual nationwide demand increase rate relative to the base year (2022) for each planning year (2030–2060). For 2030, we assume an annual nationwide demand of 12.5 PW h per year (approximately 1.36 times the 2023 levels) considering the recent rapid demand growth, which is also close to the expert assessments.^{13,14} By 2060, an economy-wide integrated model predicts China's annual demand will reach around 15.4 PW h per year,^{4,6} aligned closely with projections from Shu *et al.*¹³ and Li *et al.*¹⁴ of 15.7 PW h per year. Within these analyses, we adopt the annual electricity demand of 16.0 PW h per year in 2060 for the base scenario. Based on these assumptions, the annual demand increase rates for 2030 and 2060 serve as the basis for projecting nationwide yearly demand for the intervening planning years.

4.4 Scenario design

We design ten parallel optimization pathways toward 2060 with a 10-year step, according to the assumptions for annual emissions limitations, investment cost and electricity demand projections, biomass availability, and land availability for VRE.

Scenarios considering carbon emissions only differ in the annual emissions limitations to the base scenario (see the ESI,[†] Section S6, for emissions targets), while other input data and parameters are consistent. Definitions of these scenarios are as follows. The base scenario is the center case with target annual emissions in 2060 of about -550 MtCO_2 per year; the “NegEmis200Mt” is a scenario with more relaxed annual emissions limitations than the base scenario, targeting net-zero emissions in 2050, and slightly negative emissions (-200 MtCO_2 per year) in 2060; the “NegEmis700Mt” is a scenario with stricter annual emissions limitations than the base scenario, targeting about -400 MtCO_2 per year in 2050 and -700 MtCO_2 per year in 2060; the “NoEmisCap” scenario is designed as a control scenario without an emissions limitation for comparison.

In the “Capex1.25X” scenario, the investment costs of technologies with higher uncertainty in industry advancement are 1.25 times that of the base scenario, including wind, solar PV, CSP, pumped hydro storage, battery storage, and biomass energy with carbon capture and sequestration. For the higher electricity demand projection scenario (“Demand1.25X”), we increase the annual electricity demand by 25% compared to the base scenario from 2030 to 2060. Negative emissions technologies in power are modeled as BECCS and direct air capture with carbon sequestration in the CISPO model. To test the impacts of restricted use of BECCS, we design a limited annual biofuel availability scenario (“LimitedBECCS”, about 7.45 EJ per year) with a quarter of the base scenario, and a scenario without BECCS (“WithoutBECCS”) only using DACCS as the NET. The “VRELandOpen” and “VRELandConservative” scenarios adopt the installation capacity potential for wind and

solar PV power assessed under “Open” and “Conservative” suitable area assumptions as the upper deployable bound for each grid cell (see the ESI,[†] Section S2, for a detailed land availability assessment).

Author contributions

Ziheng Zhu: conceptualization, methodology, software, data collection and calibration, formal analysis, visualization, writing – original draft preparation; Da Zhang: conceptualization, supervision, writing – review & editing, funding acquisition; Xiaoye Zhang: supervision, funding acquisition; Xilaing Zhang: supervision, funding acquisition.

Data availability

All data used in this study are reported in the paper or ESI.[†]

Conflicts of interest

There are no conflicts of interest to declare.

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References

- 1 United Nations (UN), Statement by H.E. Xi Jinping President of the People's Republic of China at the General Debate of the 75th Session of The United Nations General Assembly, [EB/OL], 2020, https://estatements.unmeetings.org/estatements/10.0010/20200922/cVOFMr0rKnhr/qR2WoyhEseD8_en.pdf.
- 2 X. Chen, Y. Liu, Q. Wang, J. Lv, J. Wen, X. Chen, C. Kang, S. Cheng and M. B. McElroy, *Joule*, 2021, 5, 2715–2741.
- 3 Z. Zhuo, E. Du, N. Zhang, C. P. Nielsen, X. Lu, J. Xiao, J. Wu and C. Kang, *Nat. Commun.*, 2022, 13, 3172.
- 4 D. Zhang, Z. Zhu, S. Chen, C. Zhang, X. Lu, X. Zhang, X. Zhang and M. R. Davidson, *Proc. Natl. Acad. Sci. U. S. A.*, 2024, 121, e2306517121.
- 5 M. Li, R. Shan, A. Abdulla, E. Virguez and S. Gao, *Energy Environ. Sci.*, 2024, 17, 2193–2205.
- 6 X. Zhang, X. Huang, D. Zhang, Y. Geng, L. Tian, Y. Fan and W. Chen, *J. Manage. World*, 2022, 38(01), 35–66.
- 7 Ministry of Industry and Information Technology (MIIT), *xIndustrial Energy Efficiency Improvement Action Plan*, 2022, <https://www.gov.cn/zhengce/zhengceku/2022-06/29/5698410/files/e5be6f181bce4f89acb0b60d18d1acf6.pdf>.
- 8 Ministry of Industry and Information Technology (MIIT), National Development and Reform Commission (NDRC)

- and Ministry of Ecology and Environment (MEE), *Implementation Plan for Carbon Peaking in the Industrial Sector*, 2022, <https://www.gov.cn/zhengce/zhengceku/2022-08/01/5703910/files/f7edf770241a404c9bc608c051f13b45.pdf>.
- 9 The State Council of the People's Republic of China, *Action Plan for Peaking Carbon Emissions Before 2030*, 2021, https://www.gov.cn/gongbao/content/2021/content_5649731.htm.
 - 10 C. E. C. (CEC), *Analysis and Prediction Report on the National Electricity Supply and Demand Situation from 2023 to 2024*, 2024, <https://www.cec.org.cn/detail/index.html?3-330280>.
 - 11 IEA, *Electricity 2024*, Iea, Paris technical report, 2024.
 - 12 N. E. A. (NEA), *The National Energy Administration releases statistical data on the national power industry for 2023*, 2024, https://www.nea.gov.cn/2024-01/26/c_1310762246.htm.
 - 13 Y. Shu, Y. Zhao, L. Zhao, B. Qiu, M. Liu and Y. Yang, *Proc. CSEE*, 2023, **43**, 1663–1671.
 - 14 H. Li, D. Liu and D. Yao, *Proc. CSEE*, 2021, **41**, 6245–6258.
 - 15 Y. Wang, Q. Chao, L. Zhao and R. Chang, *Carbon Neutrality*, 2022, **1**, 15.
 - 16 S. Chen, X. Lu, Y. Miao, Y. Deng, C. P. Nielsen, N. Elbot, Y. Wang, K. G. Logan, M. B. McElroy and J. Hao, *Joule*, 2019, **3**, 1895–1912.
 - 17 J. Tian, S. Zhou and Y. Wang, *Environ. Impact Assess. Rev.*, 2023, **102**, 107161.
 - 18 J. A. de Chalendar and S. M. Benson, *Joule*, 2019, **3**, 1389–1393.
 - 19 J. E. Bistline and G. J. Blanford, *Proc. Natl. Acad. Sci. U. S. A.*, 2016, **113**, E3988.
 - 20 Y. Lei, Z. Wang, D. Wang, X. Zhang, H. Che, X. Yue, C. Tian, J. Zhong, L. Guo and L. Li, *et al.*, *Nat. Clim. Change*, 2023, **13**, 693–700.
 - 21 L. Liu, G. He, M. Wu, G. Liu, H. Zhang, Y. Chen, J. Shen and S. Li, *Nat. Energy*, 2023, **8**, 870–880.
 - 22 B. U. Schyska, A. Kies, M. Schlott, L. von Bremen and W. Medjroubi, *Joule*, 2021, **5**, 2606–2624.
 - 23 P. R. Brown and A. Botterud, *Joule*, 2021, **5**, 115–134.
 - 24 Y. Ou, *Nat. Energy*, 2023, **8**, 1309–1310.
 - 25 A. Millot, P. Lubello, E. M. Tennyson, M. Mutembei, M. Akute, D. Mentis, S. Pye, A. Hawkes and S. Sterl, *Cell Rep. Sustainability*, 2024, **1**, 100222.
 - 26 Y.-M. Wei, J.-N. Kang, L.-C. Liu, Q. Li, P.-T. Wang, J.-J. Hou, Q.-M. Liang, H. Liao, S.-F. Huang and B. Yu, *Nat. Clim. Change*, 2021, **11**, 112–118.
 - 27 S. Collins, J. P. Deane, K. Poncelet, E. Panos, R. C. Pietzcker, E. Delarue and B. P. Ó. Gallachóir, *Renewable Sustainable Energy Rev.*, 2017, **76**, 839–856.
 - 28 M. Li, R. Shan, A. Abdulla, E. Virguez and S. Gao, *Energy Environ. Sci.*, 2024, **17**, 2193–2205.
 - 29 X. Guo, X. Chen, X. Chen, P. Sherman, J. Wen and M. McElroy, *Nat. Commun.*, 2023, **14**, 2447.
 - 30 J.-L. Fan, Z. Li, X. Huang, K. Li, X. Zhang, X. Lu, J. Wu, K. Hubacek and B. Shen, *Nat. Commun.*, 2023, **14**, 5972.
 - 31 Y. Wang, R. Wang, K. Tanaka, P. Ciais, J. Penuelas, Y. Balkanski, J. Sardans, D. Hauglustaine, W. Liu and X. Xing, *et al.*, *Nature*, 2023, **619**, 761–767.
 - 32 S. Zhang and W. Chen, *Engineering*, 2022, **14**, 64–76.
 - 33 X. Deng, F. Teng, M. Chen, Z. Du, B. Wang, R. Li and P. Wang, *Nat. Commun.*, 2024, **15**, 1085.
 - 34 J.-L. Fan, J. Fu, X. Zhang, K. Li, W. Zhou, K. Hubacek, J. Urpelainen, S. Shen, S. Chang and S. Guo, *et al.*, *Nat. Clim. Change*, 2023, **13**, 807–815.
 - 35 D. Zhang, X.-D. Huang, J.-T. Zhong, L.-F. Guo, S.-Y. Guo, D.-Y. Wang, C.-H. Miao, X.-L. Zhang and X.-Y. Zhang, *Adv. Clim. Change Res.*, 2023, **14**, 941–951.
 - 36 China Renewable Energy Engineering Institute, *China Renewable Energy Development Report 2022 (in Chinese)*, 2023.
 - 37 M. S. Ziegler, J. M. Mueller, G. D. Pereira, J. Song, M. Ferrara, Y.-M. Chiang and J. E. Trancik, *Joule*, 2019, **3**, 2134–2153.
 - 38 T. Levin, J. Bistline, R. Sioshansi, W. J. Cole, J. Kwon, S. P. Burger, G. W. Crabtree, J. D. Jenkins, R. O'Neil and M. Korpås, *et al.*, *Nat. Energy*, 2023, **8**, 1199–1208.
 - 39 National Energy Administration (NEA), *Medium and long-term development plan of pumped storage (2021-2035) (Draft for soliciting opinions) (in Chinese)*, [EB/OL], 2021, https://m.sohu.com/a/483866703_121123711/?pvid=000115_3w_a.
 - 40 N. A. Sepulveda, J. D. Jenkins, F. J. De Sisternes and R. K. Lester, *Joule*, 2018, **2**, 2403–2420.
 - 41 X.-J. Xiao and K.-J. Jiang, *Adv. Clim. Change Res.*, 2018, **9**, 138–143.
 - 42 S. Cobo, A. Galan-Martin, V. Tulus, M. A. J. Huijbregts and G. Guillen-Gosalbez, *Nat. Commun.*, 2022, **13**, 2535.
 - 43 R. Wang, W. Cai, L. Yu, W. Li, L. Zhu, B. Cao, J. Li, J. Shen, S. Zhang and Y. Nie, *et al.*, *Sci. Data*, 2023, **10**, 384.
 - 44 J. Young, N. McQueen, C. Charalambous, S. Foteinis, O. Hawrot, M. Ojeda, H. Pilorgé, J. Andresen, P. Psarras, P. Renforth, S. Garcia and S. Mijndert van der, *One Earth*, 2023, **6**, 899–917.
 - 45 R. Wang, H. Li, W. Cai, X. Cui, S. Zhang, J. Li, Y. Weng, X. Song, B. Cao, L. Zhu, L. Yu, W. Li, L. Huang, B. Qi, W. Ma, J. Bian, J. Zhang, Y. Nie, J. Fu, J. Zhang and C. Wang, *Environ. Sci. Technol.*, 2022, **56**, 16082–16093.
 - 46 Z. Zhang, M. Chen, T. Zhong, R. Zhu, Z. Qian, F. Zhang, Y. Yang, K. Zhang, P. Santi, K. Wang, Y. Pu, L. Tian, G. Lü and J. Yan, *Nat. Commun.*, 2023, **14**, 2347.
 - 47 S. Joshi, S. Mittal, P. Holloway, P. R. Shukla, B. Ó. Gallachóir and J. Glynn, *Nat. Commun.*, 2021, **12**, 5738.
 - 48 S. Srinivasan, *Technology and Economics of Smart Grids and Sustainable Energy*, 2019, vol. 4, p. 13.
 - 49 NEA and NDRC, *Notice on Establishing a Coal Power Capacity Payment Mechanism*, 2023, https://www.ndrc.gov.cn/xxgk/zcfb/tz/202311/t20231110_1361897.html.
 - 50 P. Beiter, T. Mai, M. Mowers and J. Bistline, *Nat. Energy*, 2023, **8**, 1240–1249.
 - 51 MIT Energy Initiative and Princeton University ZERO lab, *GenX: a configurable power system capacity expansion model for studying low-carbon energy futures*, 2022, Accessed: 2024-02-25, <https://github.com/GenXProject/GenX>.
 - 52 H. Zhang, D. Zhang and X. Zhang, *Renewable Sustainable Energy Rev.*, 2023, **173**, 113080.
 - 53 J. A. Dowling, K. Z. Rinaldi, T. H. Ruggles, S. J. Davis, M. Yuan, F. Tong, N. S. Lewis and K. Caldeira, *Joule*, 2020, **4**, 1907–1928.
 - 54 Development and Reform Commission of Henan Province, *Notice on Promoting the Healthy and Sustainable Development*

- of Distributed Photovoltaic Power Generation, 2023, <https://fgw.henan.gov.cn/2023/11-02/2840540.html>.
- 55 Energy Administration of Shandong Province, *Announcement on the release of the assessment results of the carrying capacity of distributed photovoltaic access to the power grid*, 2024, <https://www.shandong.gov.cn/jpaas-jpolicy-web-server/front/info/declaresource?iid=3003055447b9442a87bb4f1d7cf8f396>.
 - 56 M. Fajardy and N. Mac Dowell, *Energy Environ. Sci.*, 2017, **10**, 1389–1426.
 - 57 L. H. Goulder, X. Long, C. Qu and D. Zhang, *China's Nationwide CO₂ Emissions Trading System: A General Equilibrium Assessment*, NBER Working Paper No. 31809, 2023, <https://www.nber.org/papers/w31809>.
 - 58 X. Han, X. Chen, M. B. McElroy, S. Liao, C. P. Nielsen and J. Wen, *Appl. Energy*, 2019, **237**, 145–154.
 - 59 R. Romero, A. Monticelli, A. Garcia and S. Haffner, *IEEE Proc. Gener. Transm. Distrib.*, 2002, **149**, 27–36.
 - 60 H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, A. Simmons, C. Soci, S. Abdalla, X. Abellan, G. Balsamo, P. Bechtold, G. Biavati, J. Bidlot, M. Bonavita, G. Chiara, P. Dahlgren, D. Dee, M. Diamantakis, R. Dragani, J. Flemming, R. Forbes, M. Fuentes, A. Geer, L. Haimberger, S. Healy, R. J. Hogan, E. Hólm, M. Janisková, S. Keeley, P. Laloyaux, P. Lopez, C. Lupu, G. Radnoti, P. Rosnay, I. Rozum, F. Vamborg, S. Villaume and J.-N. Thépaut, *Q. J. R. Meteorol. Soc.*, 2020, **146**, 1999–2049.
 - 61 N. R. E. Laboratory, *2020 Annual Technology Baseline: NREL Reference 7MW*, 2020, https://nrel.github.io/turbine-models/2020ATB_NREL_Reference_7MW_200.html.
 - 62 P. R. Brown and F. M. O'Sullivan, *Renewable Sustainable Energy Rev.*, 2020, **121**, 109594.
 - 63 S. Jerez, I. Tobin, R. Vautard, J. P. Montávez, J. M. López-Romero, F. Thais, B. Bartok, O. B. Christensen, A. Colette, M. Déqué, G. Nikulin, S. Kotlarski, E. van Meijgaard, C. Teichmann and M. Wild, *Nat. Commun.*, 2015, **6**, 10014.
 - 64 D. E. H. J. Gernaat, H. S. de Boer, V. Daioglou, S. G. Yalew, C. Müller and D. P. van Vuuren, *Nat. Clim. Change*, 2021, **11**, 119–125.
 - 65 Microsoft, *Global Building ML Footprints*, 2023, <https://github.com/microsoft/GlobalMLBuildingFootprints>.
 - 66 T. Lu, P. Sherman, X. Chen, S. Chen, X. Lu and M. McElroy, *Nat. Commun.*, 2020, **11**, 4750.
 - 67 R. Xu, Z. Zeng, M. Pan, A. D. Ziegler, J. Holden, D. V. Spracklen, L. E. Brown, X. He, D. Chen and B. Ye, *et al.*, *Nat. Water*, 2023, **1**, 113–122.
 - 68 D. E. H. J. Gernaat, P. W. Bogaart, D. P. V. Vuuren, H. Biemans and R. Niessink, *Nat. Energy*, 2017, **2**, 821–828.
 - 69 P. Lin, M. Pan, H. E. Beck, Y. Yang, D. Yamazaki, R. Frasson, C. H. David, M. Durand, T. M. Pavelsky and G. H. Allen, *et al.*, *Water Resour. Res.*, 2019, **55**, 6499–6516.
 - 70 X. Lu, L. Cao, H. Wang, W. Peng, J. Xing, S. Wang, S. Cai, B. Shen, Q. Yang and C. P. Nielsen, *et al.*, *Proc. Natl. Acad. Sci. U. S. A.*, 2019, **116**, 8206–8213.
 - 71 OpenStreetMapContributors, *OpenStreetMap data*, 2022, <https://www.openstreetmap.org>.
 - 72 Natural Earth, *1:50m Cultural Vectors*, 2023, <https://www.naturalearthdata.com/downloads/50m-cultural-vectors/>.
 - 73 Global Energy Monitor (GEM), *Renewable Energy and Other Power*, 2023, <https://globalenergymonitor.org/>.
 - 74 W. Wan, P. Döll and H. Zheng, *Water Resour. Res.*, 2022, **58**, e2022WR032380.
 - 75 National Energy Administration (NEA), *Construction and operation of photovoltaic power generation in 2022*, 2023, https://www.nea.gov.cn/2023-02/17/c_1310698128.htm.
 - 76 China Electric Power Planning and Engineering Institute (CEPPEI), *Report on China Electric Power Development 2023 (in Chinese)*, 2023.
 - 77 National Development and Reform Commission (NDRC), *Typical power load curves in each provincial power system (in Chinese)*, 2019, <https://www.ndrc.gov.cn/xxgk/zcfb/tz/201912/P020191230336066090861.pdf>.